Screening Without Selling: Employer Language and Applicant Queues in an Opaque Labor Market*

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Abstract

In models of directed search, posted wages direct applications. On Kariyer.net, Türkiye's largest job board, however, employer advertisements rarely mention the price of the job, but instead focus on the qualifications they seek. Out of 972,366 ads posted between January 2017 and January 2022, four out of five ask for experience and three out of four ask for education. Only 2.65 percent mention cash pay and only 0.43 percent state a specific amount or a range. We find that pay disclosure is persistent among employers. When an ad mentions cash pay, the employer's next ad does so 44.2 percent of the time, compared to 1.4 percent after an ad without a mention. This pattern holds across positions, ad texts and years, and stated pay amounts persist similarly. An employer's past disclosure predicts its future disclosures, controlling for the position and industry of the job. Disclosure has only a weak association with observable market conditions. Although very few ads mention pay, each ad receives on average 365 applications, six to ten times the averages reported for job boards in Chile and the United States. The number of applications tracks the labor market and the employer much better than the ad's measurable content. Applicant pools follow the stated requirements only partially. Taken together, our findings indicate that pay disclosure is a persistent employer practice, in a market where ads act as screening devices but rarely function as sales pitches.

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1 Introduction

Job advertisements can carry two types of information. Requirements tell job seekers whether they qualify for a job, and pay tells them whether they should apply. In models of directed search, posted wages steer applications, and competition for applicants through the wage can make the market efficient (Moen, 1997; Wright et al., 2021). If instead pay is bargained after matching, the standard matching model is efficient only under a special condition (Hosios, 1990). Employers' choice of whether or not to post pay can thus affect the efficiency of the market and thereby the welfare of workers and firms.

Employers may have a reason to state requirements but not pay. Screening applications is costly (Wolthoff, 2018), and requirements deter unqualified workers. Leaving pay out creates room for negotiation. In the model of Michelacci and Suarez (2006), bargaining over pay may be optimal for firms, but the equilibrium without posted pay fails to maximize total surplus, which is output net of search costs. When job seekers search for jobs without seeing pay, they might waste effort applying to jobs they wouldn't work for anyway. It is therefore important to understand how a labor market operates when postings usually list requirements but not pay.

Kariyer.net, Türkiye's largest job board, is such a market. Advertisements on Kariyer.net list requirements frequently but pay infrequently. Three-quarters of advertisements list educational requirements, and 80 percent list experience requirements. Yet, only 2.65 percent of advertisements disclose any form of pay, and 0.43 percent specify pay amounts or ranges. In comparison, between 13 and 20 percent of job postings in the United States and Chile list pay levels (Banfi and Villena-Roldán, 2019; Marinescu and Wolthoff, 2020), while in Norway, about 60 percent of advertisements contain some pay-related information (Audoly et al., 2024).

Kariyer.net advertisements also receive far more applications than advertisements on other job boards. The average posting on Kariyer.net receives 365 applications, compared to approximately 35 at trabajando.com in Chile and 59 at CareerBuilder in the U.S. (Banfi and Villena-Roldán, 2019; Marinescu and Wolthoff, 2020). Hence, job advertisements can attract hundreds of applicants even if they do not reveal the level of compensation. We study how recruitment takes place in this setting, where pay transparency is virtually absent. We investigate how applicants self-sort among available job advertisements, and how employers decide how many applications to accept and whether to disclose pay.

We find that explicit pay information is very rare, yet pay information is not an isolated characteristic of any given advertisement. Employers' behavior of reporting pay persists across jobs, ads, and years. An employer's past advertising behavior also predicts whether its

future ads report pay, over and above the characteristics of the job. Our primary contribution is demonstrating this employer-level persistence. Our evidence of persistence coexists with long applicant queues, in which only some applicants are qualified for the job. We do not attempt to test whether job requirements are substitutes for reported pay or whether the absence of pay information leads to a less efficient labor market. Instead, we just report facts that any explanation for the observed choices needs to explain.

The data include 972,366 advertisements posted between January 2017 and January 2022. For each advertisement, we observe the text of the ad as well as which fields the employer filled out from the platform's drop-down menus. The transparent rule measures what the ad text says. We constructed the transparent rule using 550 ads that were labeled by our AI labeler and verified by a human labeler on 100 of those ads. We then held the rule constant and tested it on an additional 300 ads which our AI labeler labeled without seeing any labels. We link all ads posted by the same employer using a unique, anonymous employer ID. The platform also provides us with the number of views and the number of applications for each ad. These aggregate statistics also provide the share of applicants who have each level of education, and their average age and years of experience. We never see any data on any individual job seeker. After employers close an ad, they may optionally complete a survey explaining why the ad was closed. One option is that the employer got enough applications. We call these responses closure reports. Closure reports allow us to compare the size of an employer's queue with the employer's assessment of the size of the queue.

Our first set of results concerns pay disclosure. Cash pay disclosure is persistent within employers. If an employer's ad has cash pay, then the next ad from the same employer (from any job) posted at a later date has cash pay 44.2 percent of the time; after an ad without cash pay it is 1.4 percent. This is true even if the two ads are for different jobs or have very different texts, and even if the ads are a year or more apart. The same result holds for pay amounts and minimum wage references. In two-thirds of cases where pay is mentioned in both ads, the second ad does not repeat any pay-related sentence from the first ad verbatim. So persistence is not simply from copying the same pay-related sentences.

Because most mentions of cash pay do not tell how much the cash pay is, our finding is not about how employers disclose particular pay levels. It is about how employers talk about pay in general. An employer's past practice predicts whether the employer will mention cash pay. This information on past practice predicts nearly as well as the position of the job. Combining past practice with the position and the sector leads to better predictions. The measures of market conditions we have show only weak associations with disclosure. These findings indicate that employers' recruitment strategies are persistent over time. But since we

can only observe what an employer writes, not the underlying processes that produced it, we cannot determine whether the persistence comes from preferences or habits in recruitment.

The remaining findings describe the setting in which employers decide whether to state pay. Our second set of findings relates to how many applicants there are. We define a market as a position code in a province in a given month. A decomposition within the sample shows that the market effect alone explains 54.9 percent of the variation in applications in logs. Our 26 text-based indicators add 1.4 percentage points. The employer effect, which captures the average number of applications, adds another 21 percentage points. If we enter the employer effect before the text variables, then the text variables contribute 0.3 percentage points. The number of applications tracks the market and the employer much more closely than the text variables that we measure. However, because the employer effect captures many things, including pay and reputation, we cannot identify which of these causes the association between applications and employer.

Our third set of findings analyzes who applies. We compare advertisements posted by the same employer for the same position code in the same province, which we refer to as recurring recruitment. We also control for all factors that are common across all advertisements for a particular position code and province in a given month. If an advertisement requires a university degree, the proportion of university graduates among applicants is 2.4 percentage points higher, and the proportion with a high school education or less is 2.2 percentage points lower. These patterns indicate only partial sorting because there are also many unqualified applicants. In advertisements requiring a university degree, on average 21.1 percent of applicants have no university degree. The results for experience requirements are similar. A numerical experience requirement is associated with 0.45 more years of average applicant experience, and each additional year of the required minimum is associated with 0.34 more years. Numerical experience requirements are also associated with approximately 11 percent fewer applications and fewer applications per view, which is what one would expect if some workers who do not satisfy the requirement choose not to apply. However, the job may change along with the text, so we interpret all of these results as correlations.

Fourth, closure reports provide information on the employer side of the queue. The median queue among advertisements is 214 applications. Among advertisements where the employer reports that they have received enough applications, the median is 250 applications, and a quarter are at or below 108 applications. This queue, relatively common on the platform, appears adequate for these employers.

We contribute to the literature on directed search with partial wage information. Wage posting varies across settings, and many advertisements contain little useful wage information (Brenčič, 2012; Batra et al., 2023). Worker surveys also document how often workers know

their pay in advance or bargain over it (Hall and Krueger, 2012). Even "hidden" wages are found to correlate with the number of applicants, and applicant characteristics match job requirements (Banfi and Villena-Roldán, 2019). A large portion of wage-relevant information is found in the job title (Marinescu and Wolthoff, 2020). Advertised wage levels affect job seekers' interest (Belot et al., 2022), and disclosing wages changes how applications respond to the offered pay (Balgova et al., 2025). Job postings also contain information about variation in skill requirements across firms and over time (Deming and Kahn, 2018; Hershbein and Kahn, 2018; Modestino et al., 2020). Employers also use job ads to state explicit preferences, for example for gender and age (Delgado Helleseter et al., 2020). We add to this literature by studying a context in which pay information is nearly absent. Requirements are still associated with applicant characteristics within each employer, but the number of applications is more strongly associated with the employer and the labor market than the text of the advertisement.

A different strand of the literature studies how job seekers allocate their job applications across jobs, sectors and locations (Marinescu and Rathelot, 2018; Davis and Samaniego de la Parra, 2024). Closure reports provide complementary information about employer perceptions of the size of the queue, which cannot be determined from the number of applications.

We also contribute to the literature on pay transparency. Audoly et al. (2024) classify the pay and non-pay information in Norwegian job advertisements and show, using multiple methods including a survey experiment, that these features signal employer quality. Most ads in their data contain some pay information, which contrasts with our setting. Since we have so little pay information, we cannot explore whether advertised pay is used as a signal of quality. We can instead study how employers attract workers without pay information, and which employers tend to disclose pay. Other papers examine the consequences of pay transparency on bargaining within firms and cross-firm wage comparisons (Cullen and Pakzad-Hurson, 2023; Cullen, 2024). Others study laws that require pay in job ads, in Slovakia (Škoda, 2022) and in Colorado (Arnold et al., 2025). A related literature studies the information workers can obtain about employers (Sockin and Sojourner, 2023). Prior evidence from the US already shows that the firms that post the most job ads rarely disclose wages (Batra et al., 2023). Posted wages also show that many firms set nearly the same pay for a job across locations (Hazell et al., 2022). We present additional evidence on the same employers over time, showing that cash pay references are persistent across an employer's positions, texts, and years, and that an employer's history of pay references predicts its future use.

Our analysis is descriptive. Employers select the information they share, and closure reports are submitted voluntarily. We do not see the choices made by individual applicants, nor the chain of subsequent events, from interviews and offers to confirmed hires and actual

wages, after an applicant applies. Therefore, we cannot quantify the welfare loss caused by opaque advertisements or predict the effects of a forced disclosure policy on the market. We can, however, describe the market such a policy would change.

The rest of the paper proceeds as folloes. Section 2 describes the platform and the labor market. Section 3 describes the data. Section 4 analyzes the text of job advertisements. Section 5 examines where applications flow. Section 6 characterizes the applicants themselves. Section 7 discusses the employer closure reports. Section 8 documents the persistence of cash-pay mentions across employers and over time. Section 9 describes the limitations of the data, and Section 10 concludes.

2 Setting

2.1 The platform

Kariyer.net is the largest online job board in Türkiye. The platform primarily focuses on recruiting in the formal sector for professional, technical, administrative, sales and managerial positions. In our data, 82.0 percent of advertisements are for occupations in ISCO-08 major groups 1 to 4, from managers to clerical support workers (International Labour Organization, 2012). Thus, while the data capture white-collar recruitment on the platform well, they do not reflect all jobs in Türkiye.

Employers begin by selecting the position title, province, sector, department, position level and work form from menus provided by the platform. They then write a description and the required qualifications in a free-text box. Job seekers create profiles containing information about their age, education and work history. Applying for a position on the platform is free, but employers must pay to display advertisements.

Job seekers can search by position title, location, sector and level using the menu fields. There is no menu field for pay during our sample period. Employers can enter a precise title and location through the menu fields and simply omit any mention of pay. Thus, any pay-related information in the data has to be entered into the free-text box.

2.2 Packages and recorded dates

Initially, advertisements were sold with posting windows lasting up to 60 days. Starting from June 2019, new advertisers bought 30-day packages, while firms with existing agreements kept their 60-day packages for a transition period. The data contain the first and last publication dates of an advertisement, as well as the recorded end date. The first and last publication dates differ only for advertisements posted more than once. We exclude these advertisements

from our analyses of applications and applicants, because the platform pools their applications across postings (Section 3). The length of time between initial publication and the recorded end date generally corresponds to the duration of the posting package that the employer purchased. Among advertisements posted only once, 84.9 percent have a duration of exactly 29, 30, 59 or 60 days (see Appendix Table D2). We take these durations to correspond to the posting windows. However, the durations reveal very little about the time taken to fill the vacancy.

2.3 The labor market

During the period we consider, there was high unemployment and rising inflation. The overall unemployment rate was 10.9 percent in 2017 and 10.4 percent in 2022 (although our sample ends in January of that year). Youth unemployment was 20.8 and 19.4 percent, respectively, in those years (Turkish Statistical Institute, 2018, 2023). The expansion of higher education continued as well. In the 2022–2023 academic year, 208 higher education institutions enrolled approximately 6.95 million students, with 903,673 students having graduated the previous academic year (Council of Higher Education, 2023). The OECD highlights a large informal sector, and a mismatch between the educational background of many graduates and the jobs they perform (OECD, 2023). Kariyer.net serves the formal segment of this labor market. The application data discussed later show that many of the advertised positions attract large numbers of educated applicants.

The annual rate of consumer price inflation averaged 11 percent in 2017, hit 25 percent in October 2018 and increased to 48.7 percent in January 2022 (Turkish Statistical Institute, 2022). This creates competing incentives for employers to disclose pay. An amount that employers publish can quickly become outdated, which might discourage them from posting any figure at all. On the other hand, given the pace of price changes, employers might find it valuable to share current pay information with potential applicants, which would encourage them to disclose it. Section 8 traces the evolution of pay disclosure over the sample period without trying to distinguish between these two mechanisms. The legally established minimum wage serves as a common benchmark in Turkish wage-setting practices (OECD, 2023), and some advertisements refer to this instead of specifying a concrete amount.

3 Data

3.1 The advertisements

The platform has 1,121,695 records posted between January 1, 2017, and January 31, 2022. We drop two groups of postings to prepare the analysis sample. The first group, 6,048 postings, had missing key fields or a location outside Türkiye. The second group, 143,281 observations, had posting histories that could not be treated as a single recruitment. We keep a history with r appearances when $r \leq 10$ and its recorded span is no longer than $60r + 30(r-1)$ days. The rule allows one paid posting window for each appearance and a gap between appearances. Histories that run longer, or that have more than ten appearances, are dropped. Table B1 shows each step. The analysis sample has 972,366 advertisements from 101,287 anonymized employers, with 11,183 position codes and all 81 provinces. Istanbul alone accounts for 50.9 percent of the advertisements. Ankara follows with 9.5 percent and Izmir with 6.7 percent.

We drop 6,048 postings for missing key fields or for being located outside of Türkiye. We also drop 143,281 observations for having posting histories that are not suitable to be treated as a single recruitment. A history of r appearances with recorded span no longer than $60r + 30(r-1)$ days is kept, where $r \leq 10$. The rule gives one paid posting window for each appearance and one gap between appearances. Histories that exceed the time limit or appear more than ten times are excluded. Table B1 displays the process. The remaining sample includes 972,366 ads from 101,287 anonymized employers, 11,183 occupation codes and 81 provinces. The largest number of ads, 50.9 percent, is in Istanbul; 9.5 percent are from Ankara; and 6.7 percent from Izmir.

If an ad is posted repeatedly, the platform aggregates the total number of views, applications and applicant characteristics over its entire appearance period. Since we do not know the date on which each application was submitted, we cannot decompose these three measures by appearance. Thus, any discussion of applications or applicants relies on the subset of 828,399 ads that appear only once, which we refer to as *posted-once sample*. Our exclusion criterion applies to posting histories, not to individual ads. Appendix Table B1 presents our data selection steps, starting from the initial platform records through to the final regression samples.

3.2 Applications and applicant pools

For each advertisement, we observe a) views and applications, b) the share of applicants at each education level, and c) the applicants' average age and experience. The platform

supplies these as summaries for each advertisement, and we do not observe the people behind them. Table 1 gives summary statistics. Across all advertisements, the mean number of applications is 409 and the median is 228. The average advertisement gets 3,463 views, and the mean ratio is 11.4 applications per 100 views. Posted-once advertisements have a mean queue of 365 applications and a median of 214. About 47 percent of the average applicant pool are university graduates. Averaged across advertisements, applicants are about 35 years old and have about 8.5 years of experience.

Table 1. Summary statistics

	All advertisements			Posted	Regression
	Mean	SD	Median	once, mean	sample, mean
Panel A. Advertisement records					
Recorded posting window (days)	56.7	38.6	59.0	46.2	49.0
Applications	409	924	228	365	444
Views	3,463	7,195	2,172	2,998	3,320
Applications per 100 views	11.4	6.4	10.6	11.6	13.1
Panel B. Applicant pool, platform averages per advertisement					
High school or less (%)	16.3	20.8	7.0	16.2	16.8
Associate degree graduates (%)	10.2	9.6	8.0	10.3	10.0
University graduates (%)	47.3	19.8	52.0	47.4	46.0
Postgraduates (%)	14.2	12.3	11.0	14.0	13.4
Current students (%)	12.1	8.5	11.0	12.0	13.8
Average age (years)	35.3	3.8	35.0	35.3	34.7
Average experience (years)	8.5	3.0	8.0	8.5	8.1
Panel C. Pay information in the text					
Mentions cash pay (%)	2.65	16.05	0.00	2.74	2.79
States a number (%)	0.43	6.52	0.00	0.46	0.52
Mentions a non-wage benefit (%)	3.80	19.13	0.00	3.82	4.05

Notes. The columns use all 972,366 advertisements, the 828,399 posted-once advertisements and the 225,887 advertisements of the common regression sample (Table B1). The platform reports applicant education, age and experience as averages for each advertisement, not as individual records. Current students are applicants still enrolled in an associate or university program. Applicants whose education is not recorded make up 0.02 percent of the average pool.

3.2.1 Reasons given for closing an ad

Employers might provide reasons for closing an ad, and 23.28 percent of ads contain a recorded reason (Table D1 in Appendix). Employers report finding a worker via Kariyer.net in 8.85 percent of ads, and through another channel in 3.42 percent. They report receiving enough applications in 3.98 percent, and a frozen or canceled position in 7.03 percent. There are just 19 ads with the reason "no candidate found". This low frequency does not imply that most

hirings succeed, since reporting is optional, and employers who give a reason might differ from others. The reasons are self-reported. "Found a candidate" could indicate a job offer, a hire, or an earlier stage in the process. So we treat "enough applications" as an employer's subjective assessment of the queue, and maintain the site's terminology when referring to reports that the employer found a candidate.

3.2.2 Comparison with other job boards and vacancy data

The volume of applications on Kariyer.net is high relative to other sites (Table 2). The average of 365 applications per posted-once advertisement contrasts with approximately 35 on Trabajando.com in Chile, 59 on CareerBuilder in the US, and 11 on Dice.com (Banfi and Villena-Roldán, 2019; Marinescu and Wolthoff, 2020; Davis and Samaniego de la Parra, 2024). A job board in India reports about 41 applications per advertisement, and a job board in the Chinese city of Xiamen about 63 (Chaturvedi et al., 2025; Kuhn and Shen, 2015). Longer posting windows explain only part of the gap. Under the 30-day package, the mean queue is 234 applications and the median is 143. Under the 60-day package, the mean is 460 and the median is 285 (Appendix Table D3).

Pay information is also rarer on Kariyer.net than in any other source in Table 2. Only 2.65 percent of Kariyer.net advertisements mention cash pay, and only 0.43 percent state an amount or a range. On the job boards in Chile and the US, 13 to 20 percent of advertisements state pay (Banfi and Villena-Roldán, 2019; Marinescu and Wolthoff, 2020; Davis and Samaniego de la Parra, 2024). The share is 14 percent across all online job posts in the US (Batra et al., 2023). On the job boards in Xiamen and India, about 59 percent and over 87 percent of advertisements state pay (Kuhn and Shen, 2015; Chaturvedi et al., 2025). In Norway, about 60 percent of job advertisements contain pay-related information (Audoly et al., 2024). The lowest share in the other sources is 4.4 percent, for vacancies posted in Addis Ababa (Balgova et al., 2025). Different regions and types of jobs are present on the various platforms, and a single ad might correspond to a different number of vacancies in each case. The sources also define pay information in different ways. This is merely an indication of the magnitude of the application queues and of how rarely pay is stated.

4 What Employers Write

4.1 Screening information and offer information

We split the text into two sets of categories: information about the worker and information about the job. *Screening information* is about the kind of applicant the employer is looking

Table 2. Applications and pay information across job boards and vacancy data

Platform or data	Period	Applications per advertisement	Advertisements with pay information (%)	Source
Kariyer.net, Türkiye	2017–2022	365	2.65 / 0.43	This paper
Other job boards				
Trabajando.com, Chile	2008–2014	35	13.4	Banfi and Villena-Roldán (2019)
CareerBuilder, United States	2011	59	about 20	Marinescu and Wolthoff (2020)
Dice.com, United States	2012–2017	11	17	Davis and Samaniego de la Parra (2024)
XMRC, Xiamen, China	2010	about 63	59	Kuhn and Shen (2015)
Job portal, India	2018–2020	41	over 87	Chaturvedi et al. (2025)
Other vacancy data				
Online job posts, United States	2012–2017	–	14	Batra et al. (2023)
Public vacancy database, Norway	2021–2024	–	about 60	Audoly et al. (2024)
Posted vacancies, Addis Ababa, Ethiopia	2022	–	4.4	Balgova et al. (2025)

Notes. Applications per advertisement are means. The medians are 214 for Kariyer.net and 4 for Dice.com. The Dice.com figures refer to standard postings and count both applications on the site and click-throughs to external application pages. The XMRC figure is the number of applications that a typical advertisement received. For Kariyer.net, applications refer to the 828,399 posted-once advertisements, and the pay shares refer to all 972,366 advertisements. The first Kariyer.net share counts any mention of cash pay, and the second counts an amount or a range. The other sources count a) a wage shown in the advertisement (Trabajando.com), b) a posted wage (CareerBuilder and XMRC), c) a pay level or range (Dice.com and India), d) any wage information (US online job posts), e) pay-related information (Norway), and f) explicit pay information (Addis Ababa). The CareerBuilder data cover Chicago and Washington, DC. The Addis Ababa data cover job boards, an online aggregator and newspapers. A dash means that the source reports no comparable figure. The sources differ in occupations and period, and an advertisement may stand for a different number of openings in each source.

for, such as experience, education, software and technical skills, foreign languages and place of residence. *Offer information* is about the job, such as pay, non-wage benefits, contract terms and working conditions.[1] We fixed these categories before calculating the prevalence of each category in the data. The categories define what the text says rather than what it

[1]Contract terms refer to explicit references to full-time, part-time, permanent, fixed-term, temporary, seasonal, freelance, internship, or project-based work. Non-wage benefits include compensation in kind, services, and allowances designated for a particular use, such as meal cards or meal allowances, transport or travel allowances, private health insurance, housing, a company car or phone, childcare, discounts, and training. The list of the rules gives the Turkish words for each category. Meal, transport and travel allowances count as non-wage benefits rather than cash pay.

might be used for. For example, a requirement could also indicate pay, and a description of working conditions could also serve as a screening device.

Figure 2 displays the skew of the text. The most common information is about experience, found in 80.5 percent of advertisements. The second-most common is education, in 74.8 percent. Software or technical skills appear in around half of the advertisements. Foreign languages appear in 38.1 percent and the desired location in 24.2 percent.[2] Information about the offer is far less prevalent. Only 5.4 percent of advertisements have mentions of contract terms, 3.8 percent non-wage benefits, and 2.6 percent cash pay. Working conditions appear in 30.7 percent of advertisements, but three-quarters of those descriptions are for things like travel, shift work, or overtime. Appendix Table A1 reports the details. The title of this paper reflects this difference in screening information versus offer information. It is possible that advertisements still attract workers through some channel that our measures do not capture, such as reputation.

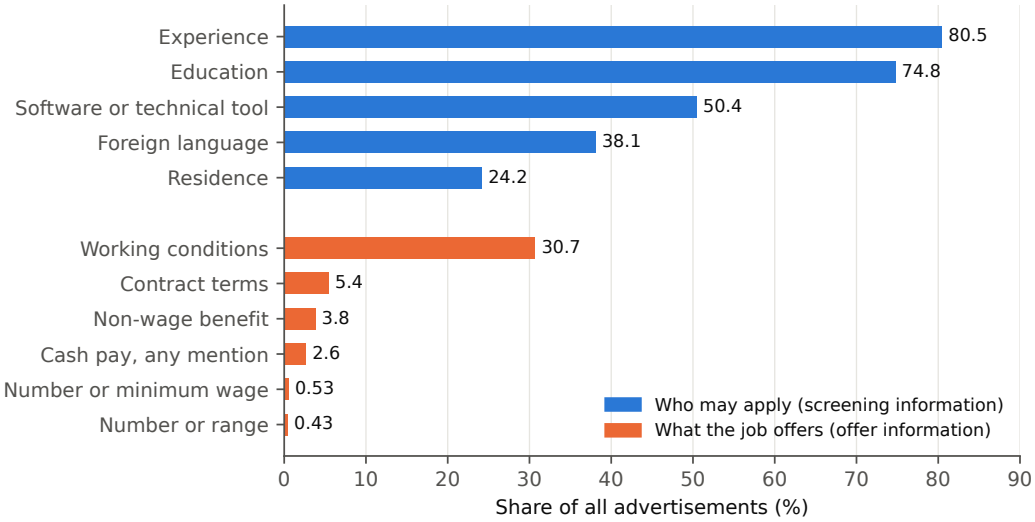


Figure 1. Information in employer-written text

Notes. All 972,366 advertisements. The figure shows the percentage of advertisements mentioning each category. Categories can overlap. "Number or minimum wage" means the advertisement includes a specific numeric amount or range, or a reference to the statutory minimum wage, or both.

The screening language can be vague. Of the advertisements that mention experience, 32.5 percent have no number of years and instead use adjectives such as *deneyimli*, meaning "experienced." Mentions of education can allow for multiple levels. Mentions of individual levels add up to 95.4 percent of advertisements, while 73.7 percent name at least one level, which amounts to an average of 1.30 levels per advertisement that mentions a level. A numeric bound on experience is usually a floor. In posted-once advertisements with a bound,

[2]For example, *Anadolu Yakası'nda ikamet eden* translates to "living on the Asian side of Istanbul".

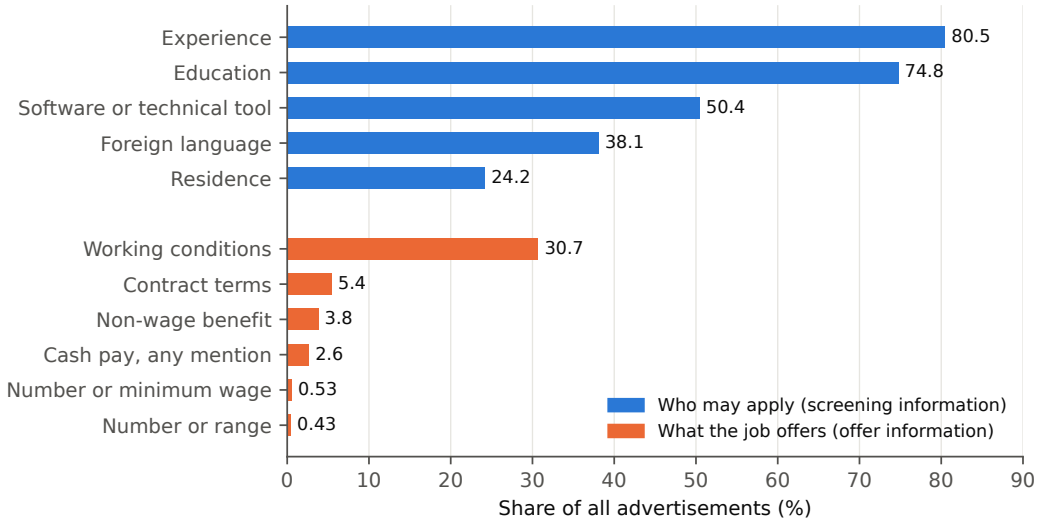


Figure 2. Information in employer-written text

Notes. All 972,366 advertisements. Each bar gives the share mentioning that category. Categories may overlap. "Number or minimum wage" includes a numerical amount or range, a reference to the statutory minimum wage, or both.

87.8 percent provide a lower bound and not an upper bound. Some advertisements indicate preferences rather than requirements, and some state that experience is not necessary.

Our measures only cover the free text, but workers have more information available. The position code, province, sector, and work form are known for more than 99.8 percent of all advertisements. The department is known for 96.1 percent and the position level for 73.3 percent. The work form could contain information that the employer does not repeat in the text, which would partly explain why so few advertisements contain contract terms. Pay is different, since there was no structured field for it during the sample period. Workers might be able to deduce what the job will pay based on the position title, the employer, or prevailing local wages. The figures report a gap in information that is explicitly stated by the employer. That does not imply that workers know nothing about how much these jobs are likely to pay.

4.2 What pay mentions say

A reference to pay often says little about the level of pay. Table 3 describes the 25,735 advertisements with a cash-pay mention. More than a quarter of them (28.9 percent) say only that the job is paid. Another 23.3 percent make a qualitative claim such as *dolgun ücret*, or "generous pay." References to premiums, bonuses, and commissions can be used to describe compensation without specifying how much money a worker should anticipate making. We classify these mentions together.

Because numbers and references to the minimum wage are more precise, we categorize those separately. A number or range appears in 4,145 advertisements, or 0.43 percent of the analysis sample, and the minimum wage is named in 1,085. Since 107 advertisements contain both, 5,123 advertisements, or 0.53 percent, give at least one of these more concrete forms of pay information. Even among cash-pay mentions the share is only 19.9 percent. The rules miss some amounts, but even when we allow for them, fewer than 1 percent of advertisements state an amount (Section 4.3). The amounts are also hard to compare, because they may be gross or net, cover different periods, or mix fixed and variable pay.

Table 3. What cash-pay mentions say

Content	Advertisements	% of all ads	% of cash-pay mentions
Any cash-pay mention	25,735	2.65	100.0
Conveys no pay level			
Generic statement that the job is paid	7,425	0.764	28.9
Premium or bonus language	8,348	0.859	32.4
Variable pay, such as premiums or commission	9,060	0.932	35.2
Qualitative claim, such as *dolgun ücret*	6,004	0.617	23.3
Fixed, base or daily pay mentioned without an amount	3,429	0.353	13.3
Pay to be discussed later	303	0.031	1.2
Negotiable pay	257	0.026	1.0
Conveys a pay level			
Number or range	4,145	0.426	16.1
Statutory minimum wage named	1,085	0.112	4.2
Number or minimum wage (either)	5,123	0.527	19.9

Notes. All advertisements, N = 972,366. One advertisement can contain several kinds of pay language, so the rows overlap. A kind of pay language counts only in a sentence or list item that the rules read as an offer of pay. Allowances for meals, transport and travel count as non-wage benefits, not as cash pay. The last row counts each advertisement once. 107 advertisements contain both a number and the minimum wage.

Explicit wage posting is far more common on the platforms studied elsewhere. In the Chilean data of Banfi and Villena-Roldán (2019), 24,867 of 184,920 cleaned advertisements display a wage, or 13.4 percent (their online appendix, Table A5). About 20 percent of advertisements in the CareerBuilder data of Marinescu and Wolthoff (2020) state a wage. The platforms are also built differently. CareerBuilder has a salary field, and Trabajando.com records an intended wage even when it is hidden from applicants. So the comparison describes the information available on Kariyer.net and says little about Türkiye as a country.

In 885 ads, or 0.09 percent, the job posting explicitly requests the candidate's expected salary. The flow of information in these ads is reversed, with the worker providing salary expectations to the employer. The specific amount given could potentially influence the

salary the employer offers to the applicant (Roussille, 2024). Because the sample is too small for analysis, we include it solely for the purpose of a control group.

4.3 How we measure the text

A deterministic program produces the text measures from named regular-expression rules. ChatGPT SOL 5.5 (OpenAI) wrote the first version of the program at our request, and Claude Opus 5.5 (Anthropic) wrote our revisions of the rules for pay, non-wage benefits, experience and education. No language model assigned any of the labels used in the analysis. We can trace each label to the rule and the passage that generated the label. The program strips HTML code, normalizes Turkish characters and spacing, and searches for the appropriate word forms. Because of Turkish suffixes, it is not reliable to simply search for the word stem; the rules specify the forms that are accepted. The rules also recognize common English and German phrases. Each pay word is evaluated in its sentence or list item. A pay word is considered a pay offer unless the surrounding words a) describe a job duty like payroll, b) are part of a legal disclaimer or c) ask for the applicant's desired salary. A number is considered a pay amount only if there are surrounding words that indicate the number is pay.

We developed the rules on a set of ads, and tested them on another. We started by coding 550 advertisements using Claude, who had not seen the labels on the rules. We modified the rules whenever they differed from Claude's code. We then held the rules constant, and drew a test sample of 300 ads. We drew the sample from the 865,158 ads of employers whose ads we had not yet read. Claude coded the sample in the same way as before. We did not have any humans code the sample, and Claude also wrote the code of the modified rules. The test therefore measures how well the rules match Claude's reading of new advertisements.

To check Claude's reading, the first author also coded 100 advertisements in the development sample, without seeing the labels or Claude's code. She and Claude give the same code in 97 percent of these advertisements, for cash pay, pay amounts, and years of experience and education (Appendix Table A3). In all six differences on pay, Claude's code follows the codebook. In the three differences on cash pay, Claude and the rules find cash pay as the codebook defines it, and the human coder does not. The three differences on pay amounts are advertisements that name the statutory minimum wage, which she counted as an amount. So the 11 advertisements that she codes with an amount are exactly the 11 that the rules flag with a pay number or the minimum wage, the measure we use in Section 8.

We measure the accuracy of the rules using two shares: precision, which is the share of the ads flagged by the rules that Claude also codes as positive, and recall, which is the share of the ads that Claude codes as positive that the rules also flag. We weight both shares

to describe the full population of 865,158 advertisements. The rules rarely flag a mention falsely, and so precision is 0.96 or higher in each category (Appendix Table A4). However, the rules do miss some mentions of pay. Recall is 0.88 for cash pay, and 0.74 for pay amounts, compared to 0.96 to 0.99 for experience and education.

We can also adjust the shares calculated by the rules to account for these errors. After adjusting, 2.94 percent of the 865,158 ads mention cash pay, compared to 2.67 percent using the rules. The share of ads with a pay amount or range is 0.56 percent (95% CI ranges from 0.43 to 0.74), compared to 0.42 percent using the rules. For experience and education, the adjustment changes the shares by at most one percentage point. Fewer than 1 percent of ads state a pay amount, even accounting for the amounts that the rules miss.

Errors in the measures could also bias the regressions in Section 6, and the fixed effects do not eliminate this bias. Errors that are not correlated with the applicant pool will bias the coefficients on requirements towards zero. The fixed effects can increase this bias, because requirements typically don't vary much in a repeated recruitment, but errors often do. The misses are also a function of the language of the advertisement, and an employer might alternate between English and Turkish in a repeated recruitment. Since the language could also influence who applies, we cannot tell the direction of this bias. Therefore, we repeat our regressions using only the Turkish-language ads (Appendix Table B10).

5 Where Applications Go

First, we examine the number of applications and explore how well the text features explain the variation in queue size. A queue is defined as the number of applications submitted to a job posting. A market refers to a position code, province, and month combination. The regressions include a fixed effect for each market, so we keep only markets with at least three posted-once advertisements.[3] In Table 4, we add sets of variables one by one to a regression of $\log(1 + \text{applications})$ and report the share of variation captured at each stage. The decomposition describes in-sample fit and is not a test of prediction.

Market effects explain 54.9 percent of the variation. Structured variables for position level and work form add just 0.3 percentage points. The 26 text indicators add 1.4 percentage points, and employer effects add a further 21.2 percentage points, yielding a final R^2 of

[3]In a market with one advertisement, the market effect fits the advertisement exactly. In a market with two advertisements, the market effect fits the average of the two advertisements, so only one comparison is left within the market. Markets with one or two advertisements would therefore raise the share of variation explained by market effects for purely mechanical reasons. These small markets would also leave little variation for the text measures to explain. The restriction keeps 413,863 of the 827,864 posted-once advertisements with a position code, in 59,472 markets. Appendix Table B6 repeats the analysis with all 827,864 advertisements. The text indicators still add far less than the employer effects.

Table 4. How much of the variation in applications each block explains

	Month markets			Quarter markets		
	R^2	Adj. R^2	Added	R^2	Adj. R^2	Added
Market effects only	0.549	0.473		0.532	0.473	
+ Structured fields (position level, work form)	0.552	0.476	0.3	0.535	0.476	0.3
+ Employer-written text	0.566	0.493	1.4	0.549	0.492	1.4
+ Employer effects	0.777	0.680	21.2	0.761	0.680	21.3
Market effects, structured fields and employer effects, no text	0.774	0.676		0.758	0.675	
Advertisements		413,863			542,985	
Markets		59,472			60,650	
Employers		66,267			77,099	

Notes. The outcome is log(1 + applications) for advertisements posted once. A market is a position code, a province and a month (or a quarter). Only markets with at least three advertisements are used. Each of the first four rows adds a block of variables to the row above it. Added is the increase in R^2 in percentage points. The fifth row leaves out the text, so the gap between the fourth and fifth rows is what the text adds when it enters last. Employer-written text is the full set of 26 text indicators of Table 6. Adjusted R^2 corrects for the number of effects and coefficients.

77.7 percent. Note that the order of the last two variable sets is important. If we put the text indicators after the employer effects, then their incremental contribution drops to only 0.3 percentage points. Adjusted R^2, which accounts for the number of parameters being estimated, yields a similar picture. Changing the definition of markets from monthly to quarterly keeps 542,985 advertisements and also has little effect. Keeping all 827,864 posted-once advertisements with a position code raises the share explained by market effects to 75.0 percent, because 300,411 advertisements are alone in their market. The text indicators then add 0.8 percentage points and employer effects add 11.5 points. So the text indicators still add far less than the employer effects (Appendix Table B6). Our 26 indicators also leave out other features of the text. When we add the length of the text and its main language, the text adds 3.2 percentage points before the employer effects and 0.8 points after them. Employer effects still add 19.9 points (Appendix Table B7).

The bulk of the variation in queue size is accounted for by market and employer effects, and very little is captured by the text features we code. This pattern is consistent with the literature showing that job titles contain useful information regarding pay and applications (Marinescu and Wolthoff, 2020), and that employer information is important for search (Sockin and Sojourner, 2023). Given that most advertisements do not disclose pay, the title and the employer may provide important clues about likely pay. However, the decomposition does not show us what workers value. Employer effects could reflect differences in pay,

benefits, employer reputation, hiring policies, the number of vacancies, and visibility on the website. Some features present in the text but absent from our indicators may be valuable to certain applicants.

Moreover, queue size is only one possible outcome. Different requirements could attract different applicant pools without changing the size of the queue. The next section looks at who applies to a job using a different metric.

6 Advertisement Content and Applicant Pools

6.1 Within-market comparisons

We consider all ads for the same position and province, and look at ads posted by the same employer. We call these *recurrences*. Ads within a recurrence need not target the same vacancy. We compare across ads within the recurrence, and control for market characteristics common to all ads in the same position-province-month market. For advertisement i, we estimate

$$Y_i = L_i'\beta + \alpha_{g(i)} + \delta_{m(i)} + X_i'\gamma + \varepsilon_i. \tag{1}$$

where Y_i is the outcome, $\alpha_{g(i)}$ is the recurring-recruitment fixed effect, and $\delta_{m(i)}$ is the market-month fixed effect. The indicators L_i include a) cash pay with and without a number, b) non-wage benefits, c) five education levels, and d) experience with and without a number. The two cash-pay categories are mutually exclusive, and each is contrasted with no cash-pay mention. X_i contains 16 other text indicators, along with work-form and position-level fields.[4]

The recruitment-group fixed effects absorb time-invariant differences between employer-position-province groups. The market-month fixed effects absorb conditions that are constant for the same position and province in a given month. We are thus left with variation in wording within recurring recruitments, after accounting for these market conditions and the other controls. Standard errors are clustered by employer and calendar month. Our main model does not control for the advertised posting window, since the window length might vary with recruitment conditions. We add these controls in Appendix Table B3. This is closer to a like-for-like comparison than one across different employers, but the advertised job could

[4]The other text indicators denote foreign languages, software, certificates, driving licenses, residence, military service, age limits, contract terms, and working conditions. They also flag general applications, mass hiring, staffing agencies, talent pools, urgent recruitment, requests for salary expectations, and pay words in other contexts (e.g., accounting duties). Men in Türkiye serve compulsory military service, and advertisements typically state whether the employer requires the male applicant to have completed service or be exempt.

still change. We thus interpret β as a set of conditional associations. These estimates do not indicate what would happen if an employer rephrased an otherwise identical advertisement.

The regressions begin with all once-posted advertisements with complete data. Ads that are unique in their recurring recruitment or in their market month provide no comparison, because the group's fixed effect absorbs the ad completely. These estimates therefore draw on the same regression sample as our primary results, which contains 225,887 advertisements from 18,594 employers with at least two ads in a recurring recruitment (Appendix Table B2). These advertisements receive a mean of 444 applications, compared with 409 in the full analysis sample and 365 among all once-posted advertisements. Thus, the estimates refer to repeat postings for the same job, which might be different from other hiring on the platform.

Pay amounts and ranges vary very little within groups. Out of 225,887 ads in the common regression sample, 1,183 report a specific pay figure. Of those, 794 belong to the 347 recurring recruitments that alternate between stating a pay figure and not doing so. These are posted by 259 employers (Table 5). Requirements differ much more. In 9,721 groups, some ads mention a university degree and some do not. This variation in support matters when we compare the precision and stability of the estimates.

Table 5. How many advertisements identify each comparison

Text indicator	Ads with a mention	Groups that switch	Mentions in those groups	Employers behind them	Within-group SD
Common sample, N = 225,887					
Cash pay, no number	5,110	1,298	2,633	965	0.073
Cash pay, number	1,183	347	794	259	0.041
Non-wage benefit	9,149	2,339	4,629	1,420	0.098
University degree	134,802	9,721	27,004	5,152	0.203
Experience, number	123,231	13,604	35,955	6,317	0.236
Same recruitment group and month, N = 64,625					
Cash pay, no number	1,535	283	367	166	0.051
Cash pay, number	519	75	100	48	0.026
Non-wage benefit	2,534	488	570	263	0.067
University degree	32,891	2,313	3,614	920	0.154
Experience, number	31,031	3,345	4,935	1,280	0.183

Notes. A group is a recurring recruitment, that is, one employer, one position code and one province. In the lower panel it is also one month. A group switches when it contains advertisements with and without the mention. Only switching groups identify the coefficients in Tables 6 to D5. The within-group SD is the standard deviation of the indicator after removing the group mean.

6.1.1 Applicant characteristics

Applicant characteristics are associated with the requirements in the text, including within recurring recruitments (Table 6). That pattern is consistent with applicants self-selecting

based on the attributes that employers mention. When an ad states a university degree, the fraction of university graduates among applicants is 2.36 percentage points higher and the fraction with a high school education or less is 2.15 points lower. A high-school mention is associated with the opposite pattern. The high-school share increases by 3.44 points and the university share falls by 3.41 points. Experience works similarly. A numerical experience requirement is associated with 0.45 more years of applicant experience on average, compared with 0.09 years for experience language without a number. Among ads with a usable stated minimum, each additional year of the minimum corresponds to 0.34 more years of applicant experience. Results are similar when weighting ads by the number of applications they receive (Appendix Table B5).

Table 6. Stated requirements and the applicant pool

	University graduates (% of applicants)	High school or less (% of applicants)	Average applicant experience (years)
University degree mentioned	2.36	−2.15	−0.017
	(0.13)	(0.15)	(0.021)
High school mentioned	−3.41	3.44	−0.053
	(0.21)	(0.27)	(0.024)
Experience, no number	−0.08	0.06	0.094
	(0.14)	(0.12)	(0.020)
Experience, with a number	−0.44	0.43	0.450
	(0.14)	(0.14)	(0.022)
Stated minimum, per year			0.340
			(0.035)
Outcome mean	46.0	16.8	8.07
Advertisements	217,567	217,567	215,735

Notes. Each column is one regression. All regressions include recruitment-group effects (employer × position × province) and market-month effects (position × province × month). They also include all five education levels, both experience indicators, the pay and benefit indicators, 16 other text indicators, work form and position level. Standard errors in parentheses are clustered by employer and by calendar month, with t critical values with 60 degrees of freedom because there are 61 months. The per-year row comes from a separate regression on the 12,729 advertisements with a usable minimum, which replaces the two experience indicators. The university column counts university graduates only. Postgraduates are a separate category.

Within-recruitment gaps are much smaller than the raw gaps. Across all once-posted ads, the share of university graduates is 17.5 points higher when a university degree is mentioned, versus 2.36 points within recurring recruitments. For the numerical experience requirement, the raw gap in average experience is 1.49 years vs 0.45 years within groups. But, as we discuss above, the raw gaps are based on all once-posted ads whereas the regression estimates are only based on ads that have been posted by the same employer for the same position in

the same province at least twice. Thus, I also compute the gaps on the regression sample itself. There the raw gaps are slightly bigger: 20.1 points and 1.69 years. The Fixed effects cut the gaps to 2.36 points and 0.43 years, and with the full set of controls to 2.36 points and 0.45 years (Appendix Table B8). So most of the raw relationship reflects differences across employers, positions and markets and not the different sample. Nonetheless, the gaps within recruitment have the same sign as the stated, intuitive criterion.

The higher share of graduates could be due to more graduates applying or fewer non-graduates applying. Since the site reports only shares, we guess the number of applicants in each group by multiplying each share by the number of applications. A university-degree mention comes with approximately 14 percent more applicants with a degree and roughly the same number of applicants without one (Table B9). Thus, a stated degree requirement is not associated with fewer applicants without a degree. On average, 21.1 percent of applicants in ads that only mention a university degree lack a degree. Therefore, for degrees the pattern is partial sorting. Ads that state a degree requirement have more graduates among their applicants, but many non-graduates still apply. A numerical experience requirement is associated with approximately 10 to 11 percent fewer applicants in both groups. Banfi and Villena-Roldán (2019) find a similar relationship between job requirements and applicant characteristics in Chile. Our comparisons condition on employer, job title and province and control for market-month effects.

Some requirements are also related to queue length (Table B4). Mentioning one education level is associated with around 11 percent more applications compared to mentioning none. In ads in Turkish, the coefficient on a university-degree mention in the application regression is lower, 0.051 compared to 0.085 for all ads (Table B10). A numerical experience requirement is associated with approximately 11 percent fewer applications and 1.12 fewer applications per 100 views. Experience language without a number seems unrelated to applications, although it appears related to applications per view. A numerical threshold may lead some workers to self-select out. We cannot separate that effect from changes in the job itself.

6.2 Pay information and the views and applications

What about the few postings that do indicate pay? We start with our main specification, which is equation (1) estimated on the common regression sample. In this specification, pay information is linked with higher application counts (Table 7). A specific pay number is linked with a coefficient of 0.166 (0.056) in the regression of $\log(1 + \text{applications})$, about an 18 percent difference for common queue sizes. A reference to a cash wage without a number is linked with a coefficient of 0.045 (0.038). This coefficient indicates a difference of around 5

percent, which is not statistically significant at the 5 percent level. A number is also linked with 0.54 (0.36) additional applications per 100 views, but this estimate is not statistically significant either. Additional benefits are linked with higher view counts (0.044 (0.022)), but not necessarily with higher application counts. View counts capture both platform visibility and worker preferences, so this comparison does not capture worker demand alone.

Table 7. Pay information, views and applications

	log(1 + applications)	log(1 + views)	Applications per 100 views
Cash pay without a number	0.045	0.008	0.272
	(0.038)	(0.029)	(0.241)
Cash pay with a number	0.166	0.096	0.536
	(0.056)	(0.045)	(0.356)
Non-wage benefit	−0.012	0.044	−0.228
	(0.040)	(0.022)	(0.144)
Outcome mean	5.373	7.628	13.13
Advertisements		225,887	

Notes. Each column is one regression on the common sample, with the effects and controls of Table 6. The two cash-pay indicators do not overlap. Both are compared with advertisements that do not mention cash pay. Views depend on search, ranking, promotion and job-seeker choice, which the data do not separate. Standard errors in parentheses are clustered by employer and by calendar month.

The numerical-pay result is less robust than the requirement results. The main estimate of 0.166 rests on the 347 recurring recruitments that switch between stating a pay number and not stating one. If we use only postings in the same month and recurring recruitment cycle, the coefficient drops to 0.122 (0.134). In that specification, cash pay without a number has a coefficient of 0.136 (0.094). But there are only 75 same-month groups that switch numerical pay, so this more restrictive comparison provides much less information (Appendix Table B3). The numerical-pay coefficient remains positive if we a) include controls for the posting window (0.170), b) exclude postings without any applications (0.163), or c) exclude mass hiring, agency, and generic job applications (0.220). The coefficient also depends on a few employers. Without the single most influential employer, the coefficient is 0.101 (0.046). Without the 5 most influential employers, it is 0.083 (0.044), which is not statistically significant at the 5 percent level. Without the 20 most influential employers, the coefficient is 0.114 (0.033). In the advertisements written in Turkish, the coefficient is 0.097 (0.049) (Appendix Table B11). So the association with applications stays positive, but its size is not stable. Without the most influential employers and in the advertisements written in Turkish, the estimates are about one half to seven tenths of the main estimate. So we put little weight on the size of these associations. None of these specifications addresses the employer's decision to disclose.

Employers might indicate a wage when they anticipate difficulty recruiting, or disclosure might correlate with other changes in the job or wage. Field experiments show that advertised wage levels affect job seekers' interest (Belot et al., 2022) and that disclosing wages changes how applications respond to the offered pay (Balgova et al., 2025). Our analysis does not allow us to estimate these effects.

7 When Employers Report Enough Applications

7.1 How large is a sufficient queue?

We have considered, thus far, only the worker side of the application market where applications go, and who applies. Closure reports provide information on the employer side. How long, in other words, is the final queue (the total number of applications the advertisement received), when an employer says it has enough applications? Table 8 displays the distribution for the 29,064 posted-once advertisements with a sufficiency report. The median is 250 applications and the lower quartile 108. The tenth percentile is 41. In this group, 23.2 percent have 100 applications or fewer, and 42.3 percent have 200 or fewer. For all posted-once advertisements, the median is 214. Hence, some employers consider queues of an ordinary size for the platform sufficient.

Table 8. Queues when employers report sufficient applications

	N	10th pct.	25th pct.	Median	75th pct.	90th pct.
All posted-once advertisements	29,064	41	108	250	519	1,000
Without mass hiring, agencies or general ads	28,108	42	109	248	512	980
Specialists and managers	12,857	43	110	237	464	812
ISCO-08 major groups 1 and 2	13,641	38	109	253	519	989
For comparison, all posted-once ads	828,399	38	95	214	431	780

Notes. The first four rows use posted-once advertisements whose employer recorded "sufficient applications" (either spelling) when closing the advertisement. The last row uses all posted-once advertisements. Queues are final application counts. Appendix Table D6 shows how many of all applications lie above the lower quartile and the median of these queues.

A possible reason for a median of 250 is that some advertisements might fill several openings simultaneously. An employer with several vacancies requires a larger number of applications than an employer with just one vacancy. However, the median hardly changes in subgroups where a single advertisement is less likely to refer to multiple openings such as 237 applications for specialists and managers or 253 for managers and professionals in ISCO-08 major groups 1 and 2 (International Labour Organization, 2012). None of these

groups ensures that there is only one vacancy per advertisement. Nevertheless, the typical sufficient queue does not appear to originate from advertisements for multiple workers.

Employers also do not just use one specific queue size. One-tenth of the advertisements with a sufficiency report ended with 41 applications or fewer and a quarter closed with 519 or more. This number is the queue size the day the employer ends the advertisement. On this day, the advertisement goes inactive on the website and no longer accepts applications. The employer may have judged the queue sufficient before closing the advertisement, and the data do not show when. So the final queue may be longer than the queue that was enough for the employer. Employers who choose this reason may differ from other employers in their hiring needs or in how they report. Other employers may require more or less.

7.2 Recorded reasons across the queue distribution

The reasons that employers record also vary with the length of the queue. In queues of less than 25 applications, two-thirds of the recorded reasons say that the position was frozen or canceled. The relationship here is almost certainly mechanical: when an employer freezes or cancels a position, the ad is closed early, stops accepting applications, and thus has a short queue. Longer queues contain a more informative story. From 75 applications and up, the leading reason recorded by employers is that they found a candidate on the platform. Its share of recorded reasons increases from 9.4 percent in the shortest queues to 49.2 percent in queues of 500–999 applications (Figure 3). So among the employers who do record a reason, those with longer queues are much more likely to report that they found a candidate on the platform.

The reports of having enough applications also follow a flatter pattern. They account for 12.0 to 17.2 percent of recorded reasons in queues of up to 999 applications, and for 23.5 percent only in queues of 1,000 or more. Thus, the employers indicate that they do have enough applications at each queue size, as we saw in Section 7.1. All of these shares refer only to employers who report a reason, which accounts for 17.9 to 29.2 percent of advertisements in each queue range (Appendix Table D4). Thus, the shares represent recorded assessments, rather than probabilities of being hired.

Pay information is not clearly linked with reports that the employer found a candidate (Appendix D). Within recurring recruitments, an advertisement that mentions cash pay without a number is 1.48 percentage points (1.17) more likely to end with any recorded reason. The advertisement is also 0.68 points (0.83) more likely to end with a report of a candidate found on the platform. Neither estimate is statistically significant. Employers who record a reason name a candidate found on the platform in 38 percent of cases. If the

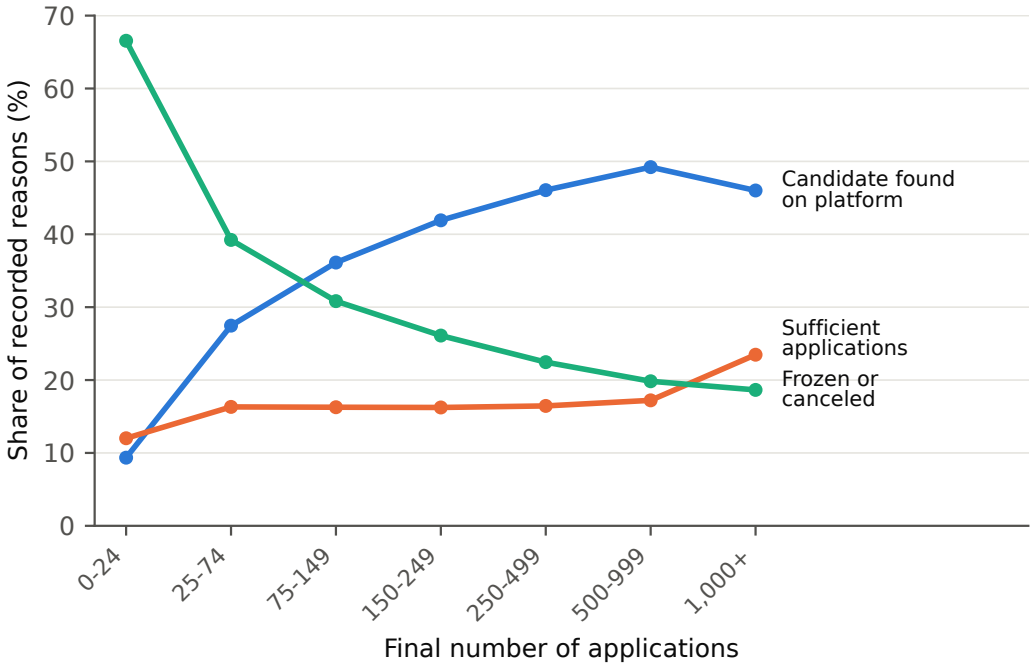


Figure 3. Recorded closure reasons by final queue size

Notes. Posted-once advertisements whose employer recorded a closure reason. The final queue is the number of applications when the advertisement closed. For each range of final queues, each line gives the share of recorded reasons that are a) a candidate found through Kariyer.net, b) enough applications or c) a frozen or canceled position. The fourth reason, a candidate found through another source, makes up the rest (Appendix Table D4). The shares describe what employers chose to record and are not hiring probabilities.

additional reports followed this pattern, they alone would raise the reports of a candidate found on the platform by about 0.57 points. So even the small difference in such reports may only reflect that these employers fill in the closure form more often.

8 Pay Disclosure across Employers and over Time

In this section we analyze which employers include pay information. We define pay disclosure as an advertisement that references cash pay, regardless of whether a monetary amount is specified. We investigate whether such disclosures are unique to individual advertisements or persist throughout an employer's hiring activities. Whether the answer is yes has implications for any regulation that mandates disclosure of remuneration in job ads, which Section 8.5 discusses. The data support disclosure as an attribute of the employer. We find a) a disclosure in one ad makes disclosure in the employer's subsequent ad likely (Section 8.1), b) the persistence characterizes stable variation across employers but does not imply causation (Section 8.2), c) correlations between disclosure and observable hiring terms are small, and

most are imprecisely estimated (Section 8.3), and d) newer employers, jobs at lower levels and certain industries have higher disclosure rates (Section 8.4).

8.1 Persistence of disclosure

To determine if disclosure persists from one ad to another, we track employers' ads sequentially. For each employer, we sort its once-posted ads by the date of their initial posting. Then we match each ad with the employer's following ad. We drop a pair when its two ads were first published on the same day, because an employer may post several near-duplicate ads at once. Table 9 and Figure 4 show how frequently the succeeding ad refers to cash wages. Disclosure is persistent. In 44.2 percent of cases, an employer's next ad discloses cash wages if the previous ad does. When the previous ad does not disclose, only 1.4 percent of the next ads do. This difference is 42.8 percentage points with a standard error of 1.1. Within recurring recruitment processes, these values are 79.1 percent and 0.7 percent. Non-wage benefits also display persistence. For consecutive ads from the same employer, the rate of benefit disclosure is 38.2 percent if benefits were mentioned previously versus 2.5 percent otherwise.

Although persistence is diminished when the two ads vary significantly, the effect remains substantial. a) In ad pairs with less than 90 percent similarity, 35.5 percent of the second ads mention cash wages after a prior disclosure, while 1.5 percent mention them after a non-disclosure. b) With ad pairs for different positions and less similar texts, 30.9 percent of second ads disclose cash wages after a previous mention and 1.4 percent after no mention. c) For ad pairs separated by one year or more, 34.4 percent of second ads disclose cash wages after a prior mention and 2.4 percent after no mention.

Persistence does not depend on vague pay language. Pay numbers and references to the minimum wage persist about as strongly as other cash-pay language. After an advertisement with a number or the minimum wage, the employer's next advertisement also states a number or the minimum wage in 39.3 percent of cases. After an advertisement without either, the share is 0.29 percent. For other cash-pay language, the rates are 39.3 and 1.2 percent (Table 10, Panel A).

Persistence also does not come only from copying the same sentence. A pay sentence is a sentence that contains a pay word. After an advertisement that mentions cash pay, the employer's next advertisement mentions cash pay in 44.2 percent of the pairs. The next advertisement mentions cash pay without repeating any pay sentence of the earlier advertisement word for word in 29.5 percent of the pairs. So two-thirds of the repeated mentions use new wording (29.5 of 44.2 percent). The other third repeat at least one pay

Table 9. Cash-pay mentions in consecutive advertisements

Pair of consecutive advertisements	Pairs	Next ad mentions cash pay (%)		Difference
		Previous ad did not	Previous ad did	
Same employer, different dates	554,515	1.39	44.19	42.81
				(1.09)
Same recurring recruitment	257,030	0.68	79.09	78.41
				(1.21)
Same employer, 365 or more days apart	21,759	2.43	34.44	32.01
				(1.68)
Same recurring recruitment, 365 or more days apart	57,452	0.80	57.86	57.06
				(1.90)
Same employer, text similarity below 90%	522,370	1.45	35.49	34.03
				(1.26)
Same employer, text similarity below 80%	508,782	1.47	32.43	30.96
				(1.35)
Same employer, other position	509,249	1.39	33.53	32.14
				(1.26)
Same employer, other position, similarity below 90%	501,259	1.41	30.95	29.54
				(1.33)

Notes. Posted-once advertisements are ordered by employer (or recurring recruitment) and first publication date. Pairs posted on the same day are dropped. A recurring recruitment is one employer, one position code and one province. Standard errors of the difference, clustered by employer, are in parentheses. Text similarity is the normalized Indel similarity of the two full texts after removing HTML, case and a leading reference number. Pay wording is kept. The rows with long gaps describe the employers who post with such gaps. They are not effects at a fixed horizon.

sentence word for word. The mentions with new wording are still far more common after a mention than after none, at 29.5 against 1.4 percent (Table 10, Panel A).

Moreover, the effect does not change over time. We divide the pairs into two periods, namely, 2017-2019 and 2020-January 2022. We keep the pair only if both advertisements are from the same period. In both periods, the next advertisement mentions cash pay in about 45 percent of cases after a mention and in 1.4 percent after none (Appendix Table C2).

Employers that disclose often in their first year also disclose often years later. For each employer, we compute the disclosure rate in its first observed year. The disclosure rate is the share of the employer's posted-once advertisements that mention cash pay. We compute the same rate for all years that are at least two calendar years after the first year. We use the 13,617 employers with at least two advertisements in the first year and at least two in

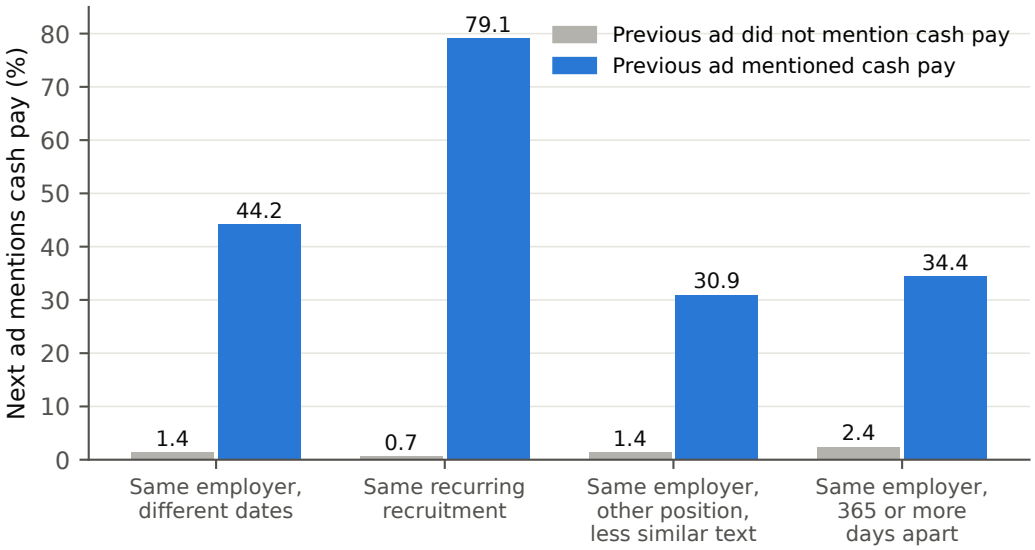


Figure 4. Cash-pay mentions in an employer's next advertisement

Notes. The figure uses consecutive pairs of posted-once advertisements and drops pairs published on the same day. Pairs are formed within employers or within recurring recruitments, as labeled. "Other position, less similar text" requires different position codes and a text similarity below 90 percent.

the later years. We regress the later rate on the first-year rate, with one observation per employer. The coefficient estimate is 0.53 (0.025). So an employer whose first-year rate is 10 percentage points higher has a later rate that is about 5.3 points higher. The data start in 2017, so an employer's first observed year may not be its first year on the platform. We therefore repeat the regression for the 3,227 employers first observed in 2018 or later. For these employers, the coefficient is 0.56 (0.045). Appendix Table C5 reports both regressions.

8.2 What persistence can and cannot show

Next, we analyze if stable differences between employers can produce the observed persistence in disclosure. Suppose that each employer decides in each advertisement whether to mention pay. Suppose also that employer f mentions pay with its own fixed probability p_f, and that each decision is independent of the employer's earlier decisions. Then, for a fixed set of pairs of advertisements by the same employer,

$$\Pr(D_t = 1 \mid D_{t-1} = 1) - \Pr(D_t = 1 \mid D_{t-1} = 0) = \frac{\mathrm{Var}(p_f)}{\bar{p}(1-\bar{p})}, \tag{2}$$

where D_t equals one if advertisement t mentions pay, and $\bar{p}$ is the average disclosure probability. The average and the variance weight each employer by its share of pairs. Appendix F derives

Table 10. Persistence of pay information and the value of employer history for prediction

A. Next advertisement of the same employer

	Pairs after a mention	After a mention (%)	After none (%)	Difference (SE)
Any cash pay	13,086	44.2	1.39	42.8 (1.1)
Pay number or minimum wage	2,440	39.3	0.29	39.0 (1.8)
Pay number only	2,026	41.9	0.23	41.7 (2.1)
Other cash-pay language	10,646	39.3	1.20	38.1 (1.0)
Any cash pay, no pay sentence repeated	13,086	29.5	1.39	28.1 (1.1)

B. Prediction of disclosure in 2021 and January 2022 from earlier advertisements

	Reduction in mean squared prediction error (%)	
Predictor	Any cash pay	Number or minimum wage
Sector	2.9	0.5
Position code	9.2	2.5
Position within sector	11.5	2.7
Employer	8.8	4.5
Position within sector, plus employer	17.4	6.0

Notes. Panel A uses consecutive pairs of posted-once advertisements by the same employer, published on different days (Table 9). The first column counts the pairs in which the earlier advertisement has the type of pay information in the row. The second column gives the share of these pairs in which the next advertisement also has that type. The third column gives the same share when the earlier advertisement does not have that type. The last column gives the difference in percentage points. Other cash-pay language is a cash-pay mention with no number and no reference to the minimum wage. A pay sentence is a sentence with a pay word. In the last row, a mention in the next advertisement counts only if none of its pay sentences appears word for word in the earlier advertisement. Standard errors clustered by employer are in parentheses. Appendix Table C3 gives more detail.

Panel B predicts whether each of the 245,034 posted-once advertisements of 2021 and January 2022 has the type of pay information in the column. Each predictor is the share of advertisements with that type in a group, such as a sector or an employer, among the 583,365 posted-once advertisements published before 2021. Each number is the percentage reduction in the mean squared prediction error, compared with predicting the overall share of the earlier advertisements for every advertisement. Position within sector uses each combination of position code and sector. The last row adds the employer's average deviation from the position-within-sector share in the earlier advertisements. Shares from groups with few advertisements are pulled toward the share of a larger group. Appendix Table C4 gives the details.

the result. The left-hand side is the gap that Table 9 reports. The right-hand side grows with the spread of disclosure probabilities across employers.

So stable differences between employers could produce the whole gap. At the observed rates, a standard deviation of about 10 percentage points in employer disclosure probabilities would produce a gap of about 43 points. The gap would arise even if each decision were independent of the last. We therefore use persistence to describe lasting differences in how employers recruit. Persistence cannot identify a behavioral habit or a causal effect of past disclosure. There are likely many causes of the persistent differences, such as firm policy and conscious decisions.

But persistence is helpful for prediction. To test this, we calculate the disclosure rate for every employer, every position, and every sector using the 583,365 advertisements that were posted only once and appeared prior to 2021. We use those rates to predict the disclosure status of the 245,034 advertisements that were posted only once during 2021 and January 2022. We assess the accuracy of each prediction using the mean squared error, which is simply the mean of the square of the difference between the predicted probability and the true value of zero or one. Using the disclosure rate specific to the employer lowers the mean squared error by 8.8 percent compared with using the overall disclosure rate. Using the rate for the specific position lowers the mean squared error by 9.2 percent. Using the rate for the specific sector lowers the mean squared error by 2.9 percent (Table 10, Panel B). Rates for each position within each sector lower the mean squared error by 11.5 percent, more than the employer rates. But the employer's past disclosure adds to them. When we add the employer's average deviation from the rates of its positions and sectors, the reduction rises to 17.4 percent. For pay numbers and the minimum wage, the corresponding reductions are 2.7 and 6.0 percent. So the employer's past disclosure adds about 6 percentage points to the reduction from the rates of positions within sectors. Thus an employer's prior disclosure is a good predictor even though it does not help us understand what is causing the difference across employers.

8.3 Disclosure and recruitment conditions

The persistent differences among employers still leave room for disclosure to vary with the recruitment conditions of each opening. Table 11 examines whether disclosure varies with the observed recruitment conditions and with the employer's own history. Each row of the table is a separate descriptive regression. Most of the estimated associations are either small or imprecisely estimated.

Inflation is an example. High inflation could prompt employers to avoid specifying a wage that might become outdated shortly. Alternatively, high inflation could make up-to-date compensation information more informative (Section 2.3). During our sample period, annual inflation was between roughly 9 and 49 percent, but the fraction of advertisements reporting cash compensation was between 1.7 and 3.2 percent per month. The regression on inflation controls for employer effects, month-of-year effects, and a linear time trend. In that regression, a 10 percentage point increase in inflation is associated with a 0.08 (0.03) percentage point increase in the share of advertisements reporting cash compensation. For pay numbers and the minimum wage, the same increase goes with a rise of 0.054 (0.019) points. So stated pay did not become rarer as inflation rose, and if anything it became slightly more common.

We find at most small associations between disclosure and market conditions (Panel A of Table 11). There is little association between disclosure and market concentration.[5] There is little association between disclosure and how long the queues are at other employers, after controlling for employer-by-year and occupation fixed effects.[6] Competing advertisements, which are advertisements by other employers for the same position in the same province and month, show a small positive association. Doubling the number of competing advertisements goes with 0.03 (0.02) to 0.04 (0.03) percentage points more disclosure, and neither estimate is statistically significant at the 5 percent level. Inflation is the only condition in Panel A with a significant association, and that association is small. To compare the conditions, Appendix Table C6 contrasts a low and a high value of each condition. The low value is the 25th percentile of the condition across advertisements, and the high value is the 75th percentile. The range between the two is called the interquartile range. The table gives the difference in disclosure between the two values, with a 95 percent confidence interval. For inflation, the low value is 12 percent and the high value is 19 percent. At 19 percent inflation, the share of advertisements that mention cash pay is 0.06 percentage points higher than at 12 percent. This difference is about 2 percent of the mean rate of 2.65 percent. For market concentration and for other employers' queues, the confidence intervals do not include differences larger than about 3 to 5 percent of the mean cash-pay rate. For competing advertisements, the intervals do not include increases larger than about 7 to 10 percent of the mean.

The employer's own recruitment experience shows no clear link either. One might expect an employer to stop stating pay once its advertisements attract enough applications. But employers disclose about as often after their first report of enough applications as before that report. The employer's own disclosure history, by contrast, remains a strong predictor of the employer's later disclosure. The comparisons in Table 11 do not rule out incentives that work at the level of the vacancy, because the measures are incomplete and the estimates do not identify causal responses. Appendix C describes each measure and gives the details of the estimates.

[5]We measure market concentration using a Herfindahl index of the share of advertisements posted by each employer in a given occupation-province-year cell, similar to the vacancy-based index of Azar et al. (2022). We use only cells with at least ten advertisements. This index measures the concentration of posting among the set of employers that do post, which differs from the concentration of jobs.

[6]The queue measure for each advertisement is the average of $\log(1 + \text{applications})$ over all advertisements posted by other employers in the same occupation, province and year. We scale the measure so that a one-unit increase means that these advertisements receive twice as many applications. Doubling these queues goes with a change of $-0.02(0.05)$ percentage points in disclosure.

Table 11. What pay disclosure does and does not track

	Coefficient	SE	Advertisements
Panel A. Market and macro conditions			
Annual CPI inflation, per 10 points	0.08	(0.03)	922,644
Effects: Employer, calendar month of year, linear trend			
Concentration of posting employers (HHI, 0 to 1)	0.62	(1.16)	824,829
Effects: Employer, occupation, year			
Other employers' queues in the same market, per doubling	−0.02	(0.05)	701,826
Effects: Employer × year, occupation			
Competing ads by other employers, per doubling	0.03	(0.02)	474,463
Effects: Employer, month			
Competing ads by other employers, per doubling	0.04	(0.03)	474,331
Effects: Employer, position, month			
Panel B. The employer's own history			
After the employer first reports sufficient applications	−0.08	(0.12)	539,016
Effects: Employer, month			
Number of earlier ads by a new employer, log	−0.07	(0.18)	78,817
Effects: Employer, market month			
Panel C. Differences across employers and jobs			
First advertisement of a new employer	3.54	(0.17)	522,760
Effects: Market month			
Number of earlier ads by a new employer, log	−0.51	(0.15)	113,679
Effects: Market month			
Lower-level job (versus manager)	2.17	(0.19)	323,420
Effects: Month			
Lower-level job, same employer	0.66	(0.12)	292,431
Effects: Employer, month			

Notes. The outcome is 100 if the advertisement mentions cash pay and 0 otherwise, so coefficients are in percentage points. Each row is one regression with the fixed effects named below it. The inflation regression clusters standard errors by calendar month. All others cluster by employer. A new employer is first seen in 2018 or later, after a full year (2017) with no advertisement, and the comparison ads come from employers first seen in 2017. Concentration is computed from employer shares of advertisements in an occupation (ISCO-08), province and year cell with at least ten advertisements. A market for the queue measure is an occupation, province and year. The queue is the average log(1 + applications) of other employers' advertisements, in units of a doubling. Competing ads are the posted-once advertisements of other employers with the same position code in the same province and month, also in units of a doubling. Lower-level jobs are new-starter, intern, staff, blue-collar and service-staff positions. Managers are middle and senior managers. All rows are descriptive associations.

8.4 Which employers and jobs disclose?

Newer employers disclose more. We call employers first observed in 2018 or later newer employers, and employers observed in 2017 established employers. The first advertisement of a newer employer is 3.5 percentage points more likely to mention cash pay than an advertisement by an established employer in the same position-province-month market (Table 11, Panel C). The later advertisements of newer employers also disclose more. The gap between newer and established employers fits differences between types of employers better than a decline in disclosure caused by experience on the platform. Among newer employers, employers

with more earlier advertisements disclose less. But within the same employer, the number of earlier advertisements shows no clear relationship with disclosure. An employer's own first-year disclosure rate also predicts its later rate better than the disclosure rate of other employers in its early markets does (Appendix Table C5).

Job level explains some of the differences. We classify five lower-level position levels: (a) new starters, (b) interns, (c) staff, (d) blue-collar workers and (e) service staff. Lower-level position advertisements are 2.2 percentage points more likely to mention cash pay than advertisements for middle and senior managers in the same month. Within the same employer, the difference is 0.7 points. Panel C of Table 11 shows both estimates.

We might think that it is easy to disclose pay in positions that pay close to the statutory minimum wage, because the employer could just state the minimum wage. However, out of 53,482 advertisements for lower-level positions that mention high school but not university education, only 0.61 percent disclose a specific amount or the minimum wage. The proportion is close to the 0.53 percent of the full analysis sample (Appendix Table E2). The 53,482 advertisements are a proxy for the type of job based on job characteristics, and we do not know if the job pays the minimum wage.

There is wide variation across sectors. In call centers, 20.0 percent of advertisements disclose cash pay, and in real estate 19.2 percent. In beauty and wellness, 13.1 percent of advertisements disclose cash pay. Less than 1 percent of advertisements in hospitals, cement producers and holding companies disclose cash pay (Appendix Table E1). Commission pay or standard pay scales could explain some of the sectoral differences, but we do not have the information to test these hypotheses.

8.5 Interpretation

A number of reasons could account for persistent heterogeneity in disclosure rates among employers. Employers might want to a) preserve bargaining power until they have learned something about the candidate's productivity or outside option (Michelacci and Suarez, 2006; Hall and Krueger, 2012), b) conceal their pay from other employers competing for the same pool of workers, c) adhere to company or industry practices that view pay as confidential, or d) recycle existing application templates. The degree of competition for candidates could play a role. An established firm might attract plenty of applications even if it does not disclose pay, consistent with the pattern that younger firms tend to be more likely to disclose (Section 8.4). A specific figure for a salary may generate expectations that a set of generic job requirements does not. Research on wage transparency has shown that disclosure can alter the bargaining dynamics between employers and employees (Cullen and Pakzad-Hurson,

2023; Cullen, 2024). We cannot distinguish these factors using our data. Nor do we have data to determine whether any of them is sufficient to explain why employers rarely disclose.

While our findings rule out the possibility that the likelihood of disclosure is determined entirely by the characteristics of a single job posting, they do not rule out the possibility that certain job postings are more likely to reveal pay. We see a high degree of correlation between disclosure decisions made by the same employer for different positions, job descriptions, and time periods, while the relationships between disclosure rates and observable market characteristics are either small or statistically insignificant. Consequently, a mandate to disclose pay would apply to a market in which employers' disclosure rates remain highly dispersed. This is not a reason to believe that regulation should be aimed at employers rather than job postings, nor is it a reason to believe that employers would ignore a regulation at the level of the job posting. Rather, it reflects questions of regulatory design.

9 Limitations

One major concern is that employers decide what information they include. Even with fixed effects, the estimates identify a fairly narrow local comparison, but the models cannot control for all possible job characteristics and cannot identify the effect of changes in specific word choices. The coefficients on the job characteristics and pay variables measure within-recruitment associations. The estimates of persistence are similarly restricted; they cannot disentangle habit formation, persistent employer characteristics, or strategic behavior.

We also face limitations in the construction of the variables. The text-matching rules used to identify the variables are simple and replicable, and they correctly identify job requirements with a low error rate. However, the rules fail to identify a non-trivial fraction of job characteristics. For cash pay, the rules also misidentify some mentions of pay that are not offers of pay. Relative to the coding by an LLM on our test sample, the rules identify a) 88 percent of the mentions of cash pay, b) 74 percent of the instances of actual pay amounts, c) 98 percent of the mentions of experience, and d) 99 percent of the mentions of education. Hence, our measures undercount pay and pay amounts. The text-matching rules were constructed using common expressions in both English and German, but the rules misclassify a larger share of mentions of job characteristics in ads written in German, which comprise just 0.25 percent of all ads. The estimates of pay numbers in the ads written in Turkish are around 60 percent of the main estimate. Our checks of the rules also have limitations. We drew our test sample after we froze the rules. But Claude coded our test sample, and Claude also wrote the code of the revisions. Claude's labels are one interpretation by an AI model. However, a human coder who checked a quarter of our development sample agreed with

Claude on 97 percent of the ads in each of four categories: a) cash pay, b) pay amounts, c) years of experience, and d) education. In all six differences on pay, Claude's code follows the codebook.

Closure reasons are self-reported by employers and aren't actual hires. Application counts are final totals without arrival dates. We don't know how many positions most ads cover, so we can't calculate applications per position or determine applications submitted after jobs were filled. We only see averages for each ad, so we can't trace individual job-seekers' trajectories or see how job-seeker skills are distributed within any queue. Our descriptions of sorting are based solely on the average for each ad. Interviews, job offers, realized wages and match quality are all unobserved.

Finally, there are issues of coverage. Kariyer.net primarily covers formal, skilled recruitment, and about half the ads are from Istanbul. Our recurring-employer regressions are based on employers who advertised repeatedly. Our findings may not apply to informal labor, to less-skilled hiring in general, or to employers who hardly ever advertise. Within these limitations, our paper offers a descriptive portrait of a black-boxed labor market. We present a) what employers tell us, b) the applicant pools they encounter, c) how employers evaluate their queues, and d) the persistence of disclosure.

10 Conclusion

Job advertisements on Kariyer.net describe a desired worker with considerable specificity, yet almost never reveal expected earnings. Four out of five ads specify a preferred amount of experience. Only 2.65 percent mention pay explicitly, and fewer than 1 percent provide a specific figure, a range, or a reference to the legal minimum. Pay mentioned explicitly is much less common than in job boards analyzed in other countries, while applicant queues are much longer than in Chile or the U.S.

The applicant characteristics are correlated with the stated requirements, but imperfectly. Within a given recruitment, the applicants are more educated when the advertisement requests more education, and more experienced when the advertisement requests more experience. Still, many applicants do not meet the stated education requirement. In advertisements that only ask for a university degree, on average 21.1 percent of the applicants have no university degree. Application volume is a different matter. After accounting for market and employer, the text features we measure add very little to understanding how many people will apply. In a labor market so opaque, the employer and the job title may give applicants useful hints about pay. The employer effects may also capture a lot of other things.

The closure reports add the employer perspective. When the employers report enough applicants, the median final queue is 250. Very large queues and very little explicit pay information coexist. The queues and the missing pay information might be connected if workers who cannot compare offers apply to many jobs, but our data does not allow us to observe this connection. The reports tell us how employers assess their pool. Whether there was a good match or whether workers had sufficient information to choose well is a separate question.

Cash pay is strongly associated with the employer's prior practice, including across different positions, different texts, and long time periods. Most of the mentions of cash pay do not say what the level of pay is, so the persistence pertains to pay language generally. Most of the correlations with measurable recruitment conditions are small and/or noisy. There could be persistent differences between employers that would lead to this pattern of results with no causal link from one decision to the next.

The main policy question is whether voluntary recruitment provides workers with the information they need. Our results help frame that question but don't resolve it. Even with enough applications, workers may be searching with very little explicit information about pay. To find out if more disclosure leads to better matches or less search, we would need exogenous variation in disclosure, along with data on later stages of the hiring process. It might be possible to get such variation by changing platform design or regulation, if that change can be credibly compared with a counterfactual. The results described here characterize the practices and the application patterns those interventions would influence.

Declarations

Data availability. The advertisement data belong to Kariyer.net and cannot be shared by the authors. Researchers can request access to the same data from Kariyer.net. The replication package contains the code, text rules, data documentation, aggregate results, and instructions. The package also contains a) the validation codebook, b) the designs of the development and test samples, c) Claude's labels for these samples and d) the human coder's labels, with the labels identified by advertisement IDs, and the record of the rules that we froze before we drew the test sample. With an authorized copy of these data, the package reproduces the tables and figures. The public inflation series is included.

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Conflict of interest. İlknur Köseoğlu Sarı is employed by Kariyer.net, which provided in-kind access to anonymized data under a research agreement. Selcen Çakır declares no conflict of interest.

Ethics. The study uses anonymized advertisement records and advertisement-level averages of applicant characteristics. It uses no individual job-seeker records and involves no contact with participants. No institutional review board (IRB) review was needed, because the study uses no data on identifiable individuals.

Use of AI tools. AI assistants (ChatGPT, Codex, and Claude) were used to assist with drafting and editing the text and with writing and checking analysis code. ChatGPT wrote the first version of the program that produces the text measures. Claude coded the validation samples and wrote the code of the revisions of the rules (Section 4.3 and Appendix A). The authors checked the analyses and take full responsibility for the paper.

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Online Appendix

Screening Without Selling: Employer Language and Applicant Queues in an Opaque Labor Market

This appendix describes the text measures, regression samples and checks, disclosure over time and across recruitment conditions, closure records, and differences across sectors and position levels. The final section derives the two identities used in the paper.

A Text Measures

How the program works

The program reads the job description and requirements written by the employer. It removes HTML code, converts text to lower case, and removes extra spaces. For the main lexicon, Turkish letters ç, ğ, ı, ö, ş, and ü are replaced by c, g, i, o, s, and u. This allows a rule to match a word whether or not the employer used Turkish characters. The program then applies named regular-expression rules and combines the matches into the categories in Table A1. The rules specify accepted inflections because short Turkish stems can also match unrelated words.

Pay requires particular care. The rules judge each pay word within its clause, which is a sentence or a list item of the normalized text. A pay word counts as an offer of pay unless the nearby words a) form a legal notice, b) describe a duty such as payroll or pay administration or c) ask for the applicant's expected salary. In advertisements that mention payroll or accounting duties, a general pay word also needs a nearby word of offer, such as *verilir* (is given). *Prim* can mean either a bonus or a social-security contribution. When the surrounding words refer to social security, insurance or premium administration, the mention does not count as offered pay. *Gündelik* counts as pay only when it means a daily wage. The rules read Turkish pay words with suffixes, pay words run together with the previous word, guaranteed earnings, and common English and German pay words. Requests for an applicant's expected salary form a separate category.

A number counts as a pay amount only if nearby words mark it as pay. The rules accept five forms of amounts. These are a) an amount with a currency, such as TL, near a direct pay term such as *maaş* (salary) or *ücret* (wage), b) a number directly beside a pay term, c) an amount with a currency followed by another part of pay, as in *1600 TL + SGK* (social security), d) an amount with a currency next to a period of pay, such as *aylık* (monthly), and e) a range followed by *arası maaş* (salary between). An amount in a legal notice, a description

of duties or a question about the applicant's expected salary does not count. Free-standing numbers do not qualify. These restrictions are intended to exclude reference numbers, years, law numbers, training hours, sales targets, and headcounts. The minimum-wage rule matches *asgari ücret*, the Turkish term for the statutory minimum wage. Within the advertisements with cash pay, the rules sort the pay words into the kinds of Table 3. A kind counts only in a clause that the rules read as an offer of pay.

Non-wage benefits follow the codebook. Allowances for meals, transport and travel count as non-wage benefits, not as cash pay. Social security and the minimum living allowance do not count as benefits, because the law requires them. Words that describe the employer's business, such as a catering or shuttle-service firm, do not count either. Training counts only when it is offered to the worker, not when the worker must give it to others.

Education and experience are measured with an extension of the rules that retains Turkish characters and recognizes common inflections, including *deneyimli* and *tecrübeli*, both meaning "experienced." A numerical statement such as *en az 3 yıl deneyimli* ("at least three years of experience") always counts as an experience mention. The five education levels are high school, associate degree, university degree, master's degree, and doctorate. An advertisement may mention several levels. Our revisions add to these rules in five ways. a) They decode HTML codes, such as the code for a space, and turn the Turkish capital İ into a plain lowercase i. b) They read more inflected forms of the two words for experience. c) They allow up to four words between a number of years and the word for experience, and they count a number of years worked in a role. d) They read *yüksek okul* (vocational college) written as two words, primary and lower secondary school, university students, and a field of study followed by *mezun* (graduate). e) They read common English and German phrases for experience and education.

A count of common English, German and Turkish words assigns each advertisement its main language. By this count, 89.9 percent of all advertisements are written mainly in Turkish, 9.9 percent in English and 0.25 percent in German.

Who wrote the program

ChatGPT (OpenAI) wrote the first version of the program at our request in August 2026. We asked ChatGPT to extract all relevant information from each advertisement and to document the method. We did not record the model version, and the chat history that we recovered does not show the version either. The first version is version 1.3.5 of kariyer_text_pipeline.py. Its run record dates its labels to 6 August 2026 and lists 972,366 advertisements. A separate lexicon file stores its 104 rules, and these rules are identical to the rules inside the program.

The extension for education and experience came later, and its code is part of the replication package. Claude (Anthropic), model claude-opus-5-5, wrote our revisions of the rules for pay, non-wage benefits, experience and education at our request in September 2026. For experience and education, the revisions add rules to those of the first version and its extension. For pay and benefits, the revisions judge each clause on its own, as described above, while the first version uses the whole advertisement as the context of a pay word. When no clause of an advertisement contains a pay or benefit word that the revised rules read, the label of the first version stands, because some of its rules span the end of a clause. The rules for the other categories are those of the first version. The program, the extension and the revisions use only general Python tools for text and data, such as regular expressions and the pandas library. They contain no call to a language model and no trained statistical classifier. So the code alone assigned every label. In November 2024, for an earlier version of this project, ChatGPT also extracted the sentences around the word *ücret* from a file of advertisements. None of the measures in this paper comes from that step.

Development samples

We developed the rules on two samples that Claude coded, an earlier sample of 150 advertisements and a development sample of 400 advertisements. We then tested the final rules on a third sample, the test sample.

For the earlier sample, we drew 1,000 advertisements with a fixed seed from 20 strata defined by the labels of the first version of the program, and then 150 of these, spread across the strata roughly in proportion to the sample of 1,000. Claude coded the 150 advertisements for every category in Figure 2, without seeing the automated labels or the strata. The weights of the earlier sample are very uneven. A coded advertisement in the education stratum stands for about 28,000 advertisements. So we use the earlier sample only to develop the rules and to check categories that the development sample leaves out.

The sampling frame of the development sample holds the 971,366 advertisements that remain after the 1,000 advertisements of the earlier sample are removed. The development sample has two parts, 300 single advertisements and 50 pairs of advertisements. We drew both parts with a fixed seed.

The single advertisements come from strata defined by the labels of the first version. Each advertisement belongs to the first stratum that fits. The strata are the 20 strata of the earlier sample and two new strata. One of the old strata, unpaid positions, is empty, because all five unpaid positions fell in the earlier sample. So 21 strata hold advertisements. The two new strata hold advertisements that the first version does not flag for pay at all, although the text

contains a pay cue. In one, a number stands next to a currency word or sign. In the other, the text contains another pay word, such as a German or English pay word or a Turkish pay word with a suffix that the first version does not list. The small strata receive a fixed number of advertisements, from 3 to 20. The seven large strata hold 92 percent of the frame. These strata share 199 advertisements in proportion to their size times a measure of how much Claude and the first version disagreed in each stratum in the earlier sample. The measure is the square root of the sum, over the categories of the earlier sample, of the variance of a disagreement. This rule is the Neyman allocation. The weight of a coded advertisement is the number of advertisements in its stratum divided by the number coded there. A coded advertisement in a large stratum stands for 4,000 to 4,800 advertisements.

The 50 pairs come from the 554,515 pairs of consecutive advertisements in Table 9. In 20 pairs, the first version flags cash pay in the first advertisement only, and in 20 pairs in the second advertisement only. In 10 pairs, it flags both advertisements. An employer appears in at most one pair, and no advertisement of a pair is among the single advertisements.

Claude (Anthropic), model claude-opus-5-5, coded the 400 advertisements our request. The coding sheet shows only a sample number, the job title and the text. The sheet shows no automated label, stratum, weight, pair, employer or date. The advertisements appear in random order, and the two advertisements of a pair are at least 20 rows apart. A script removed HTML code and masked e-mail addresses, web addresses and telephone numbers before Claude read the texts. The sheet has seven items, which cover nine categories of Table A2. The pay item has three codes, for no pay, pay without an amount and pay with an amount. So one item gives both cash pay and pay amounts. The experience item works in the same way. The definitions follow the validation codebook. Eight separate Claude sessions each coded 50 advertisements with the same written instructions, and each session could read only its own advertisements. A ninth session coordinated the work. This session read the notes of the eight sessions and changed 16 codes, so that the eight sessions applied the definitions in the same way. All 16 changes concern non-wage benefits, working conditions or general or mass hiring, and none concerns pay, experience or education. The coordinating session had designed the sample, so it knew the strata, but it made these changes without looking at the automated labels. Codes marked as unsure are left out of their category.

We revised the rules for pay, non-wage benefits, experience and education where they disagreed with Claude's codes in these two samples. Table A2 compares the final rules with Claude's codes for the 300 single advertisements of the development sample. Precision and recall are weighted with the design weights, so they describe all advertisements of the frame. For pay and benefits, the final rules agree with Claude's codes on every coded advertisement of the development sample, because we revised the rules until they did. For experience and

education, weighted precision and recall are 0.97 or higher. So these results overstate the accuracy of the rules on new advertisements. The test sample below measures that accuracy. We did not revise the rules for contract terms, working conditions and general or mass hiring. For contract terms and working conditions, precision is 0.94 or higher, but recall is only 0.75 and 0.64. Claude counts stated working hours and days as working conditions, which the rules do not read. For general or mass hiring, precision and recall are both about 0.23. The rules flag plural nouns such as *elemanlar* (staff), which the codebook does not count as a sign of several hires.

In the earlier sample, the final rules find 18 of the 25 benefit mentions that Claude codes, and Claude agrees with 18 of the 23 benefit flags of the rules. We also revised the rules for benefits on this sample. So non-wage benefits are the least reliable of the pay and benefit measures, and the test sample does not check them.

The earlier sample also covers categories that the development sample leaves out. Foreign languages agree almost perfectly. The codebook separates a mention of experience from a requirement. Only 3 of the 116 advertisements that the rules flag for experience say that applicants without experience may also apply. So the experience measure can be read as a requirement, as in Section 6.1.1. The rules flag 25 coded advertisements for male-specific wording, and Claude codes none of these advertisements as male-specific. All 25 mention military service. Military service is compulsory only for men, so such a clause concerns male applicants without excluding women. Claude codes 44 advertisements as mentioning military service, and the rules flag 19 of these advertisements. The rules flag 13 advertisements for female-specific wording, and Claude agrees on 5. We do not analyze the gender flags. Military service and general or mass hiring also serve as controls in Section 6.1, and both controls are measured with error. The other text controls in Section 6.1 were not checked. These controls flag a) certificates, b) driving licenses, c) age limits, d) staffing agencies, e) talent pools, f) urgent recruitment and g) pay words used in other contexts, such as accounting duties.

The human coder

The first author coded part of the development sample as a second, human coder. She coded the same cleaned texts with the same items and definitions as Claude, on a coding sheet that showed neither the automated labels nor Claude's codes. Claude's written instructions gave more examples of the same definitions. She coded the first 103 advertisements in the order of the sheet, which is random, except three advertisements in German, which she does not read. So she coded 100 advertisements, of which 72 are single advertisements and 28 belong to employer pairs. Claude's codes for these advertisements come from the first three of its

eight sessions, mostly the first two. She coded pay, experience, education and general or mass hiring for all 100 advertisements and non-wage benefits for 99 of them. She also coded working conditions for 12 advertisements and contract terms for 11. Table A3 reports pay, experience, education, non-wage benefits and general or mass hiring, and the replication files report the other items. Her codes give a second, independent reading of these advertisements. We compare Claude's codes with her codes, and we do not adjudicate the differences between the two coders.

Across the five categories of pay, experience and education in Table A3, the two coders give the same code in 95 percent of the 498 comparisons. For a) cash pay, b) pay amounts, c) a number of years of experience and d) education, the two coders agree on 97 percent of the advertisements, and Cohen's kappa is between 0.83 and 0.94. In the 28 advertisements of employer pairs that she coded, she agrees with Claude on cash pay in 27. For non-wage benefits, the two coders agree on 95 percent of the advertisements, and Cohen's kappa is 0.74. For general or mass hiring, they agree on 97 percent, but the human coder codes no advertisement as general or mass hiring, and Claude codes three.

Many differences between the two coders follow from how each coder applied a definition. In all six differences on pay, Claude's code follows the codebook. a) The three differences on cash pay concern a promise of fair pay in a company profile, an offer of bonuses and extra income during a training program, and a base salary plus commission stated in English. The codebook counts such statements as cash pay, and the human coder did not. b) All three differences on pay amounts concern advertisements that name the statutory minimum wage. The codebook counts these as pay without an amount, but the human coder counted them as amounts. c) In all 13 differences on any mention of experience, Claude finds a mention that the human coder did not code. About half of these advertisements ask for experience only as a preference, with *tercihen* (preferably), or use the general word *deneyimli* (experienced). The codebook counts both as a mention. d) For a number of years of experience and for education, the coders differ in three advertisements each. In two of the education cases, the human coder infers a degree from the job title, such as midwife or lawyer, although the text names no level.

The final rules agree with the human codes on 97 percent of these advertisements for cash pay and for pay amounts, on 96 percent for a number of years of experience and on 95 percent for education. We revised the rules on these advertisements, so these shares overstate how well the rules agree with a human reader on new advertisements. The human check does not cover the advertisements in German.

Test sample

We froze the rules and drew the test sample afterward. A freeze record lists the checksum of every file that defines the labels, and the replication code checks the files against the record. The frame leaves out the 1,400 advertisements of the development sample and of the sample of 1,000 from which the earlier sample was drawn. It also leaves out every advertisement of the 465 employers whose advertisements we read while developing the rules. These employers posted 11 percent of all advertisements. So the frame holds 865,158 advertisements. A check of the texts finds that no test advertisement repeats the text of an advertisement that we read. The shares of the rules in the frame are close to their shares among all advertisements.

The test sample has two parts (Table A4). The first part has 220 single advertisements from 11 strata. Two strata hold the advertisements in German, with and without a cash-pay label. Three strata look for errors in cash pay. They hold a) advertisements in which the first version of the program flags cash pay and the final rules do not, b) advertisements with a number next to a currency sign and c) advertisements with another pay word. The other strata are defined by the labels of the final rules for pay, experience and education. Weights restore the composition of the frame. The second part has 40 pairs of consecutive posted-once advertisements by the same employer, from the four cells of the cash-pay label and built as in Table 9. In 12 pairs, only the first advertisement has the label, and in 12 pairs only the second has it. In 8 pairs both have it, and in 8 pairs neither has it. An employer appears in at most one pair.

Claude, the model that coded the development sample, coded the 300 advertisements in six separate sessions of 50 advertisements. The texts were cleaned and masked as in the development sample, and each session saw only an advertisement number, the job title and the text. Each session coded pay, experience and education, with the definitions of the development sample. The logs of the sessions show that each session read only its own advertisements. No code was changed after the sessions wrote it. Claude marked 6 of the 900 codes as unsure, and these codes are left out. Of the 300 advertisements, 240 are written mainly in Turkish, 26 in English and 34 in German.

Results of the test

The rules rarely flag a mention wrongly, but they miss some mentions of pay, especially pay amounts (Panel A of Table A4). For cash pay, weighted precision is 0.97 and recall is 0.88. For pay amounts, precision is 0.98 and recall is 0.74. For any mention of experience, precision and recall are both 0.98. For a number of years of experience, both are 0.96. For education, precision is 1.00 and recall is 0.99.

Most errors for cash pay are of a few kinds. All 6 false flags among the 300 advertisements use a pay word for a job task, such as payroll, attorney fees, salary garnishment or insurance premiums. Of the 7 misses, 2 use German pay words that the rules do not list, and 3 state a bonus or an amount in phrasings that the rules do not read. In the other 2, the pay information appears together with an allowance or a legal entitlement, which the rules do not count as pay.

The rules miss more mentions in German than in Turkish or English. Advertisements in German are 0.25 percent of all advertisements, and no human coder read them. Claude finds cash pay in 19 of the 34 advertisements in German. The rules flag 17 of these 19, and one advertisement without pay. The rules find 13 of the 18 education mentions and 17 of the 23 experience mentions in German, and they flag experience wrongly in 4 advertisements in German. In the 26 advertisements in English, the rules miss no cash-pay mention and flag one advertisement wrongly.

The corrected share in Table A4 starts from the exact share of the rules among the 865,158 advertisements of the frame. The corrected share then adds the weighted mean difference between Claude's code and the label of the rules in the coded sample, computed within strata. The 95 percent interval comes from 2,000 bootstrap draws within strata. The corrected share of cash pay is 2.94 percent (2.55 to 3.53), against 2.67 percent with the rules. One advertisement accounts for most of the difference. It lists a bonus and a premium among its benefits, and it stands for 2,211 advertisements of the frame. The corrected share of pay amounts is 0.56 percent (0.43 to 0.74), against 0.42 percent with the rules. For a number of years of experience, the corrected share is 54.4 percent, against 54.1 percent with the rules. For education, the corrected share is 75.0 percent, against 74.0 percent. So the rules understate cash pay and pay amounts somewhat, but even the upper end of the interval for pay amounts is well below 1 percent of advertisements.

Claude also wrote the code of the revisions, and no human coded the test sample. So the test shows that the rules agree with Claude's reading of new advertisements. The human check of the development sample shows that Claude's reading agrees with a human reader on 97 percent of the advertisements for cash pay, pay amounts, a number of years of experience and education. The human check does not cover advertisements in German.

Errors that repeat within employers

Errors that repeat within employers could create false persistence in Section 8, for example if the rules misread a sentence that an employer reuses. The 40 pairs of the test sample test for such errors (Panel B of Table A4). Claude's codes confirm the cell of the rules in

36 of the 40 pairs. In 3 of the 12 pairs in which the rules record a mention followed by none, the first advertisement uses a pay word for a job task, so Claude finds no mention in either advertisement. In 1 of the 12 pairs with no mention followed by a mention, the second advertisement also uses a pay word for a job task. Claude confirms all 8 pairs with two mentions and all 8 pairs without a mention.

Weighted to the 504,795 pairs of the frame, Claude's codes give rates of 50.6 percent after a mention and 1.2 percent after none. The rules give 43.5 and 1.4 percent in the same frame. The pairs without a flag carry a large weight, and 8 coded pairs cannot show rare misses in them. Suppose instead that each advertisement of these pairs hides a mention at the rate that the single advertisements show for advertisements without a cash-pay flag, 0.37 percent. Then the rates are 42.9 and 1.6 percent. The gap between the two rates is then 41.3 points, against 42.1 points with the rules. This second calculation treats every hidden mention as a one-off. If hidden mentions repeat within employers, for example because an employer writes in German, persistence is higher than this calculation shows. So the errors of the rules do not appear to inflate persistence. False flags make a mention look like a one-off, so if anything they make persistence look weaker. Pay sentences that employers reuse word for word do not explain all of the persistence either (Section 8.1).

Reproducibility

Two computational checks reproduced the saved labels of the first version. Rerunning version 1.3.5 on 200 seeded advertisements, split equally between those with and without a cash-pay label of the first version, reproduced seven measures exactly. A separate comparison on 3,000 random advertisements found exact agreement between the chunked requirement code and the archived code of the extension for education and experience. The replication code also checks that the files of the rules equal the files of the freeze record, and that the labels of the test sample equal the labels that the frozen rules give. These checks show that the labels can be rebuilt, but not that the labels are right. Three checks address accuracy, a) the test sample, b) the development samples and c) the human check. The archive contains the sampling code, the codebook, Claude's codes and the human coder's codes with advertisement identifiers, the code of the rules and the code of the comparisons. For the test sample, it contains the freeze record, the code, the design and Claude's codes. The advertisement texts stay with Kariyer.net.

Table A1. Information in employer-written text

	All advertisements		Posted once	
Category	N	%	N	%
Screening information				
Experience, any mention	782,420	80.47	664,358	80.20
of which with a number	528,063	54.31	444,555	53.66
of which without a number	254,357	26.16	219,803	26.53
Education	726,882	74.75	610,286	73.67
Software or technical tool	490,533	50.45	414,015	49.98
Foreign language	370,306	38.08	305,760	36.91
Residence	235,109	24.18	200,541	24.21
Offer information				
Working conditions (mostly obligations)	298,479	30.70	252,695	30.50
Contract terms	52,899	5.44	45,766	5.52
Non-wage benefit	36,997	3.80	31,666	3.82
Cash pay, any mention	25,735	2.65	22,710	2.74
of which a number or range	4,145	0.43	3,784	0.46
of which the statutory minimum wage	1,085	0.11	1,016	0.12

Notes. The sample has 972,366 advertisements, of which 828,399 were posted once. Categories overlap. A mention need not be a binding requirement. Numerical experience is counted inside any experience. Working conditions are mostly obligations, such as shifts or travel.

Table A2. The final rules compared with Claude's coding of the development sample

	Flagged among 300 single ads						Share (%)	
Category	Rules	Claude	Both	Precision	Recall	Kappa	Rules	Corrected
Revised after reading this sample								
Cash pay	41	41	41	1.00 (1.00, 1.00)	1.00 (1.00, 1.00)	1.00	2.63	2.63 (2.63, 2.63)
Pay amount or range	16	16	16	1.00 (1.00, 1.00)	1.00 (1.00, 1.00)	1.00	0.42	0.42 (0.42, 0.42)
Non-wage benefits	29	29	29	1.00 (1.00, 1.00)	1.00 (1.00, 1.00)	1.00	3.79	3.79 (3.79, 3.79)
Experience, any mention	232	228	220	0.97 (0.95, 0.99)	0.97 (0.94, 0.99)	0.81	80.5	80.8 (78.0, 83.6)
Experience, number of years	152	154	151	1.00 (1.00, 1.00)	0.99 (0.97, 1.00)	0.97	54.3	54.9 (54.3, 56.0)
Education, any level	206	213	206	1.00 (1.00, 1.00)	0.97 (0.94, 0.99)	0.94	74.8	77.5 (75.7, 79.6)
Not revised								
Contract terms	16	27	15	0.99 (0.94, 1.00)	0.75 (0.53, 0.94)	0.68	5.44	6.81 (5.66, 8.34)
Working conditions	76	128	72	0.94 (0.87, 0.99)	0.64 (0.54, 0.73)	0.57	30.7	44.0 (39.2, 49.1)
General or mass hiring	13	14	4	0.23 (0.03, 0.57)	0.23 (0.03, 0.54)	0.26	3.23	3.29 (0.63, 6.05)

Notes. The development sample of Appendix A. We revised the rules for pay, non-wage benefits, experience and education after reading this sample, so for these categories the table overstates the accuracy of the rules on new advertisements. Table A4 tests these rules on the test sample. We did not revise the rules for contract terms, working conditions and general or mass hiring. Rules, Claude and Both count the 300 single advertisements flagged by the rules, by Claude and by both. Precision and recall are weighted with the design weights, so they describe all advertisements of the frame. Cohen's kappa is not weighted. The 95 percent intervals in parentheses come from 2,000 bootstrap draws within strata. The rule share is exact for the 971,366 advertisements of the frame, and the corrected share adds the weighted difference between Claude and the rules. Negative lower bounds are set to zero. Codes that Claude marked as unsure are left out.

Table A3. The human coder compared with Claude

Category	Advertisements	Human positives	Claude positives	Agree (%)	Kappa
Cash pay	100	23	26	97	0.92
Pay amount or range	100	11	8	97	0.83
Experience, any mention	99	62	75	87	0.70
Experience, number of years	99	45	46	97	0.94
Education, any level	100	62	61	97	0.94
Non-wage benefits	99	8	13	95	0.74
General or mass hiring	99	0	3	97	0.00

Notes. The human coder is the first author. She coded the first 103 advertisements of the random coding order of the development sample, except three advertisements in German, without seeing the automated labels or Claude's codes. She coded pay, experience, education and general or mass hiring for all 100 advertisements and non-wage benefits for 99 of them. Of the 100 advertisements, 72 are single advertisements and 28 belong to employer pairs. Codes that Claude marked as unsure are left out. Agree is the percentage of advertisements on which the two coders give the same code, and Cohen's kappa is the agreement beyond chance. The human coder codes no advertisement as general or mass hiring, so kappa is zero for that category although the coders agree on 97 percent of the advertisements. The human coder also coded working conditions for 12 advertisements and contract terms for 11, which the replication files report.

Table A4. The final rules on the test sample

A. The 220 single advertisements, weighted to the 865,158 advertisements of the frame

	Flagged					Share of the frame (%)	
Category	Rules	Claude	Both	Precision	Recall	Rules	Corrected
Cash pay	63	68	61	0.97	0.88	2.67	2.94
				(0.91, 1.00)	(0.74, 0.98)		(2.55, 3.53)
Pay amount or range	23	27	22	0.98	0.74	0.42	0.56
				(0.93, 1.00)	(0.55, 0.96)		(0.43, 0.74)
Experience, any mention	145	141	133	0.98	0.98	80.4	80.4
				(0.95, 1.00)	(0.96, 1.00)		(77.7, 82.7)
Experience, number of years	73	78	71	0.96	0.96	54.1	54.4
				(0.89, 1.00)	(0.91, 1.00)		(49.1, 58.5)
Education, any level	126	136	126	1.00	0.99	74.0	75.0
				(1.00, 1.00)	(0.97, 1.00)		(74.1, 76.0)

B. The 40 employer pairs, weighted to the 504,795 pairs of the frame

	Next advertisement mentions cash pay (%)		
Labels	After a mention	After none	Difference
Rules	43.5	1.36	42.1
Claude's codes	50.6	1.24	49.4
	(43.5, 60.6)	(1.01, 1.36)	(42.4, 59.4)
Claude's codes, misses added	42.9	1.61	41.3
	(33.5, 54.9)	(1.17, 2.23)	(31.4, 53.5)

Cell of the rules	Pairs in the frame	Coded pairs	Confirmed by Claude
Mention, then none	6,497	12	9
None, then mention	6,695	12	11
Mention in both	4,999	8	8
Mention in neither	486,604	8	8

Notes. We froze the rules before we drew the test sample. The frame holds all advertisements except the 1,400 advertisements of the development sample and of the sample of 1,000 from which the earlier sample was drawn, and every advertisement of the 465 employers whose advertisements we read while developing the rules. Claude coded the 300 advertisements in six separate sessions, without seeing any label, stratum, weight or pair. Panel A uses the single advertisements, drawn from 11 strata defined by the main language, the labels of the rules and pay words in the text. Rules, Claude and Both count the coded advertisements flagged by the rules, by Claude and by both. Precision is the weighted share of rule flags that Claude also codes as positive, and recall is the weighted share of Claude's positives that the rules also flag. The weights restore the composition of the frame. The corrected share starts from the exact share of the rules in the frame and adds the weighted mean difference between Claude's code and the rule label within strata. Codes that Claude marked as unsure are left out, 2 for pay and 1 each for experience and education. Panel B uses pairs of consecutive posted-once advertisements by the same employer, drawn from the four cells of the cash-pay label of the rules and built as in Table 9. The row with misses added replaces the pairs without a flag by pairs in which each advertisement hides a mention independently, at the rate that Panel A finds among single advertisements without a cash-pay label (0.37 percent). The 95 percent intervals in parentheses come from 2,000 bootstrap draws within strata or cells.

B Regression Samples and Robustness

Table B1 shows how we build the analysis sample. Table B2 reports each regression sample. The starting sample has 827,864 posted-once advertisements with a position code. We keep complete cases for the outcomes and controls of each specification. The fixed effects fit an advertisement that is alone in any fixed-effect group exactly. So the estimation command leaves such an advertisement out of the reported sample, and the command repeats this step until no advertisement is alone in its group. Leaving these advertisements out changes no coefficient, and it avoids standard errors that can be too small (Correia, 2015). The common sample contains 225,887 advertisements from 18,594 employers. These advertisements have a mean queue of 444, compared with 409 in the full analysis sample and 365 among all posted-once advertisements (Table 1).

Standard errors are clustered by employer and calendar month. We use the small-sample corrections and t critical values with 60 degrees of freedom described in the main text, reflecting the 61 calendar-month clusters.

Table B3 reports four checks of the pay–application associations. The first adds controls for the recorded posting window. The second absorbs one effect for each recurring recruitment and month, restricting the comparison to advertisements posted by the same employer for the same position and province in the same month. The third excludes zero-application advertisements and uses log applications. The fourth excludes mass hiring, staffing agencies, and general applications. The development sample of Appendix A shows that the indicator for general or mass hiring agrees poorly with Claude's reading, so the fourth check removes only some of the advertisements that hire several workers.

With window controls, without zeros, and without the mass-hiring group, the coefficient for cash pay without a number remains between 0.03 and 0.05. The numerical-pay coefficient remains between 0.16 and 0.22. The same-month comparison is different: the coefficients are 0.136 (standard error 0.094) without a number and 0.122 (0.134) with a number. Only 75 same-month groups switch numerical-pay status (Table 5). The larger standard error and narrow support make this an inconclusive comparison of the two forms of pay information.

Table B4 replaces the five education-level indicators with indicators for one level and for two or more levels. Both are associated with more applications than no education mention. Numerical experience requirements are associated with fewer applications and a lower application-to-view ratio. Experience language without a number is associated with a lower ratio, but not a clear difference in counts.

Table B5 weights advertisements by application counts, giving more weight to pools containing more applications rather than treating every advertisement equally. The estimates

remain close to those in the main text. For example, the university-degree coefficient for the university-graduate share is 2.41 points with weights and 2.36 points without. Because workers may apply more than once, the weighted estimates should not be read as averages over distinct job seekers.

Table B6 repeats the decomposition of Table 4 without the restriction to markets with at least three posted-once advertisements. Only the 535 posted-once advertisements without a position code are left out, because they cannot be placed in a market. Of the remaining 827,864 advertisements, 300,411 are alone in their month market, and 113,590 share their market with one other advertisement. These advertisements receive fewer applications on average, 324 and 352, against 397 in markets with at least three advertisements. The market effect fits an advertisement that is alone in its market exactly. So market effects now explain 75.0 percent of the variation, against 54.9 percent in Table 4. The text indicators add 0.8 points instead of 1.4, because the text can only explain variation within a market, and an advertisement that is alone in its market has none. Employer effects add 11.5 points instead of 21.2. Adjusted R^2, which accounts for the number of market effects, changes much less. The text adds 1.5 points of adjusted R^2 here, against 1.6 points in Table 4. With quarter markets, the text adds 0.9 points and employer effects add 14.3 points. Dropping only the advertisements that are alone in their month market keeps 527,453 advertisements, and the text then adds 1.2 points. So the restriction changes the size of each block, but not the comparison between the blocks. In every version, the 26 text indicators add at most 1.4 points of R^2, far less than the employer effects.

Table B7 adds two features of the text that our 26 indicators leave out, its length and its main language. Length enters as ten groups of equal size by the number of words. Language comes from the word count of Appendix A. Length and language alone add 2.7 points of R^2 to the structured fields, more than the 26 indicators. Together with the indicators, they add 3.2 points before the employer effects and 0.8 points after them. Employer effects still add 19.9 points. Length and language may describe the type of job or employer as much as information that applicants read.

Table B8 compares raw and adjusted gaps on the same sample. The raw gaps on the regression sample are somewhat larger than the raw gaps on all posted-once advertisements. The recurring-recruitment and market-month effects remove most of each gap. Adding the text indicators and the other controls moves the education estimate by less than 0.3 points and the experience estimate by less than 0.1 years.

Table B9 approximates the number of applicants in each education group, as Section 6.1.1 describes. The platform reports the shares as rounded percentages, so the counts are approximate. We use the logarithm of one plus each count.

Table B10 repeats the main regressions on the advertisements written mainly in Turkish, which make up 86 percent of the common sample. The language comes from the word count of Appendix A. In these advertisements, a university-degree mention goes with a university-graduate share 2.63 points higher, against 2.36 points in all advertisements. A number of years of experience goes with 0.44 more years of applicant experience, against 0.45 years. The coefficient on a university-degree mention in the regression for applications is 0.051, against 0.085 in all advertisements. The coefficient on pay amounts in the regression for applications is 0.097 (0.049), against 0.166 (0.056). The coefficient on cash pay without a number is 0.062 (0.037), against 0.045 (0.038). The coefficient on non-wage benefits is 0.050 (0.031), against −0.012 (0.040).

Table B11 shows how much the coefficient on pay numbers depends on single employers and on language. The influence of an employer is the change in the coefficient when we drop the employer's advertisements. After the fixed effects and controls are partialled out, one employer accounts for 19 percent of the covariance between pay numbers and applications that identifies the coefficient. Half of the identifying variation in pay numbers comes from 60 employers. Without the most influential employer, the coefficient is 0.101 (0.046), and without the 20 most influential employers it is 0.114 (0.033). In the advertisements written in Turkish, the coefficient is 0.097 (0.049). So the positive association does not rest on one employer, but it is smaller than the main estimate.

Table B1. Sample construction

Step	Advertisements
Records in the platform export, January 2017 to January 2022	1,121,695
Removed: missing key fields or a location outside Türkiye	−6,048
Removed: posting histories that cannot be read as one recruitment	−143,281
Analysis extract (all text measures use this denominator)	972,366
Posted once (all outcome analyses start here)	828,399
Removed: missing position code	−535
Removed: missing outcomes or controls	0
Removed: alone in their recurring recruitment or market month (repeatedly)	−601,977
Common regression sample (applications and reported outcomes)	225,887
Education-composition sample	217,567
Experience-and-age sample	215,735

Notes. The first two removals were made when the extract was created, with the two rules stated in Section 3. The code for these removals is not in our archive, so the replication package starts from the extract and checks it against a checksum. A history with r appearances counts as one recruitment when its recorded span is at most $60r + 30(r - 1)$ days and $r \leq 10$. The regression samples drop advertisements that are alone in their recruitment group or market month, repeatedly, until none is left. The common sample covers 18,594 employers and all 61 months.

Table B2. Regression samples

Model	Eligible	Complete	Singletons removed	Final N	Employers
Common sample (applications and reports)	827,864	827,864	601,977	225,887	18,594
Education composition	827,864	813,626	596,059	217,567	18,451
Experience and age	827,864	808,167	592,432	215,735	18,319
Same group and month	827,864	827,864	763,239	64,625	5,000
Stated minimum experience	107,726	104,424	91,695	12,729	2,109

Notes. Eligible advertisements are posted once and have a position code. The last row also needs a usable stated minimum between 0 and 40 years. Complete cases have every outcome and control of the model. Singleton removal runs over all fixed-effect dimensions until no singleton is left.

Table B3. Sensitivity of the application associations

Specification	Cash pay, no number	Cash pay, number	Non-wage benefit	N
Main specification	0.045	0.166	−0.012	225,887
	(0.038)	(0.056)	(0.040)	
Adds recorded-window controls	0.046	0.170	−0.013	225,887
	(0.038)	(0.058)	(0.040)	
Same group and month	0.136	0.122	−0.012	64,625
	(0.094)	(0.134)	(0.084)	
log(applications), zeros dropped	0.053	0.163	−0.010	217,567
	(0.035)	(0.054)	(0.026)	
Drops mass hiring, agencies and general ads	0.030	0.220	−0.014	213,054
	(0.033)	(0.059)	(0.041)	

Notes. The outcome is log(1 + applications) except in the fourth row. Effects, controls and clustering follow Table 6. The same-month row absorbs one effect for each employer, position, province and month. Window controls are log(1 + recorded span) and indicators for 29 or 30 and 59 or 60 days.

Table B4. How precise requirements relate to applications

	log(1 + applications)	Applications per 100 views
One education level	0.100	0.459
	(0.014)	(0.076)
Two or more education levels	0.094	0.544
	(0.019)	(0.100)
Experience, no number	0.008	−0.362
	(0.020)	(0.078)
Experience, with a number	−0.113	−1.118
	(0.018)	(0.096)

Notes. N = 225,887. Effects, controls and clustering follow Table 6, except that two indicators for the number of education levels replace the five level indicators.

Table B5. The applicant pool, weighted by applications

	University graduates (% of applicants)	High school or less (% of applicants)	Average applicant experience (years)
Panel A. Each advertisement counts once, as in Table 6			
University degree mentioned	2.36	−2.15	−0.017
	(0.13)	(0.15)	(0.021)
High school mentioned	−3.41	3.44	−0.053
	(0.21)	(0.27)	(0.024)
Experience, no number	−0.08	0.06	0.094
	(0.14)	(0.12)	(0.020)
Experience, with a number	−0.44	0.43	0.450
	(0.14)	(0.14)	(0.022)
Outcome mean	46.0	16.8	8.07
Panel B. Each advertisement is weighted by its number of applications			
University degree mentioned	2.41	−2.10	−0.012
	(0.18)	(0.22)	(0.028)
High school mentioned	−3.22	3.12	−0.014
	(0.23)	(0.29)	(0.024)
Experience, no number	0.05	0.13	0.098
	(0.13)	(0.16)	(0.020)
Experience, with a number	−0.20	0.37	0.434
	(0.15)	(0.16)	(0.024)
Outcome mean	45.5	16.3	7.54

Notes. Each column within a panel is one regression. Effects, controls, samples and clustering follow Table 6. Panel B weights each advertisement by its number of applications, so it describes the average applicant rather than the average advertisement. Outcome means in Panel B are weighted in the same way. Advertisements with no recorded applications get no weight, so Panel B uses 217,567 advertisements for the education columns and 215,685 for the experience column.

Table B6. How much of the variation in applications each block explains, all posted-once advertisements

	Month markets			Quarter markets		
	R^2	Adj. R^2	Added	R^2	Adj. R^2	Added
Market effects only	0.750	0.496		0.683	0.503	
+ Structured fields (position level, work form)	0.751	0.499	0.1	0.684	0.505	0.2
+ Employer-written text	0.759	0.514	0.8	0.693	0.520	0.9
+ Employer effects	0.874	0.665	11.5	0.837	0.685	14.3
Market effects, structured fields and employer effects, no text	0.872	0.661		0.835	0.681	
Advertisements		827,864			827,864	
alone in their market		300,411			192,831	
in a market with one other advertisement		113,590			92,048	
Markets		416,678			299,505	
Employers		98,910			98,910	

Notes. The outcome is log(1 + applications) for advertisements posted once. A market is a position code, a province and a month (or a quarter). Unlike Table 4, the table keeps every posted-once advertisement with a position code, including markets with one or two advertisements. The market effect fits an advertisement that is alone in its market exactly, so these advertisements raise R^2 in every row and leave nothing for the other blocks to explain within their market. Each of the first four rows adds a block of variables to the row above it. Added is the increase in R^2 in percentage points. The fifth row leaves out the text, so the gap between the fourth and fifth rows is what the text adds when it enters last. Adjusted R^2 corrects for the number of effects and coefficients. The numbers describe fit, not causal importance.

Table B7. How much of the variation in applications the text explains, with its length and language

	R^2	Adj. R^2	Added
Market effects only	0.549	0.473	
+ Structured fields (position level, work form)	0.552	0.476	0.3
+ 26 text indicators	0.566	0.493	1.4
+ Length and language of the text	0.584	0.514	1.8
+ Employer effects	0.783	0.688	19.9
Other orders			
Structured fields, length and language, no text indicators	0.578	0.507	2.7
Structured fields and employer effects, no text	0.774	0.676	
+ 26 text indicators	0.777	0.680	0.3
+ 26 text indicators, length and language	0.783	0.688	0.8
Advertisements		413,863	

Notes. The sample and the first three rows are those of Table 4 with month markets. Length is ten groups of equal size by the number of words in the employer-written text. Language is the main language of the text from a count of common English, German and Turkish words (90.2 percent Turkish, 9.5 percent English and 0.3 percent German in this sample). Added is the increase in R^2 in percentage points over the row above. In the lower panel, it is the increase over the structured fields for the row without text indicators, and over the row without text for the last two rows. All numbers describe in-sample fit, not prediction or causal importance.

Table B8. Raw and adjusted gaps on the same sample

	Raw gap		Regression sample		
Outcome and indicator	All posted-once ads	Regression sample	Fixed effects	Plus text indicators	All controls
University-graduate share, university degree mentioned	17.51	20.11	2.36	2.63	2.36
			(0.15)	(0.13)	(0.13)
Applicant experience, number of years stated	1.49	1.69	0.43	0.51	0.45
			(0.02)	(0.03)	(0.02)

Notes. The raw gap is the difference in mean outcomes between advertisements with and without the indicator. Fixed effects regresses the outcome on the indicator alone with recurring-recruitment and market-month effects. Plus text indicators adds the indicators of pay, benefits, education and experience, and All controls is the specification of Table 6. All columns except the first use the regression sample of Table 6. Standard errors clustered by employer and calendar month are in parentheses.

Table B9. Requirements and the number of applicants in each education group

	log(1 + applicants)		
Indicator	With a degree	Without a degree	All
University degree	0.131	−0.012	0.072
	(0.012)	(0.012)	(0.012)
High school	−0.081	0.118	0.023
	(0.018)	(0.018)	(0.017)
Associate degree	−0.031	0.110	−0.001
	(0.021)	(0.027)	(0.023)
Number of years of experience	−0.111	−0.118	−0.108
	(0.019)	(0.017)	(0.017)
Advertisements	217,567	217,567	217,567
Applicants below the stated level, advertisements that name one level (%)			
Only a university degree (369,730 ads)	mean 21.1, weighted by applications 20.2, median 16.2		
Only an associate degree (6,790 ads)	mean 19.8, weighted by applications 16.9, median 15.3		

Notes. The upper panel is one regression of equation (1) for each outcome, with the controls and fixed effects of Table 6. Applicants with a degree are the number of applications times the share of university graduates and holders of higher degrees. Applicants without a degree are the number of applications times the share with high school or less, vocational college, or university study in progress. The platform reports the shares as rounded percentages, so the counts are approximate. The lower panel uses all posted-once advertisements with at least one application that name exactly one education level. Below the level means without a university degree for the first row and with high school or less or vocational study in progress for the second. Standard errors clustered by employer and calendar month are in parentheses.

Table B10. Main estimates in all advertisements and in advertisements written in Turkish

Outcome and indicator	All advertisements	Written in Turkish
log(1 + *applications*)		
Pay amount or range	0.166	0.097
	(0.056)	(0.049)
Cash pay, no number	0.045	0.062
	(0.038)	(0.037)
Non-wage benefits	−0.012	0.050
	(0.040)	(0.031)
University degree	0.085	0.051
	(0.014)	(0.015)
Number of years of experience	−0.112	−0.097
	(0.018)	(0.019)
Advertisements	225,887	193,823
University-graduate share (points)		
University degree	2.36	2.63
	(0.13)	(0.15)
High school	−3.41	−3.31
	(0.21)	(0.20)
Advertisements	217,567	189,576
High-school-or-less share (points)		
University degree	−2.15	−2.52
	(0.15)	(0.18)
High school	3.44	3.38
	(0.27)	(0.26)
Advertisements	217,567	189,576
Applicant experience (years)		
Number of years of experience	0.45	0.44
	(0.02)	(0.02)
Experience without a number	0.09	0.09
	(0.02)	(0.02)
Advertisements	215,735	188,014

Notes. Each column of a panel is one regression of equation (1) with the controls and fixed effects of Tables 7 and 6, on the regression sample that the column implies. A count of common English, German and Turkish words assigns each text its main language. Standard errors clustered by employer and calendar month are in parentheses.

Table B11. How much the pay-number estimate depends on single employers and on language

	Pay amount or range		Cash pay, no number		
Sample	Coef.	(SE)	Coef.	(SE)	Ads
Main estimate	0.166	(0.056)	0.045	(0.038)	225,887
Without the 1 most influential employer	0.101	(0.046)	0.040	(0.038)	225,416
Without the 5 most influential employers	0.083	(0.044)	0.029	(0.037)	224,894
Without the 10 most influential employers	0.110	(0.040)	0.042	(0.035)	224,289
Without the 20 most influential employers	0.114	(0.033)	0.036	(0.036)	222,675
Advertisements written in Turkish	0.097	(0.049)	0.062	(0.037)	193,823

Notes. The outcome is $\log(1 + \text{applications})$ in equation (1) with the controls and fixed effects of Table 7. The influence of an employer is the change in the pay-number coefficient when its advertisements are dropped, computed from the partialled-out regression. The rows drop the employers with the largest influence in either direction and re-estimate the model. The single most influential employer accounts for 19 percent of the covariance between the partialled-out pay-number indicator and applications that identifies the coefficient. Half of the identifying variation in the pay-number indicator comes from 60 employers. Standard errors clustered by employer and calendar month are in parentheses.

C Pay Disclosure over Time and across Employers

Table C1 reports disclosure by year, with January 2022 shown separately. Cash-pay mentions remain between 2.38 and 2.86 percent. Numerical pay rises from 0.32 percent in 2017 to between 0.39 and 0.49 percent in later periods. These small changes leave explicit pay rare throughout the sample. In particular, the rate is 0.44 percent in 2021 and 0.39 percent in January 2022; the annual data do not establish a response to the late-2021 inflation rise.

Table C3 splits the pairs of Table 9 by the type of pay information. Pay numbers and references to the minimum wage persist about as strongly as other cash-pay language across all of an employer's advertisements, and slightly less strongly within recurring recruitments. A pay sentence is a sentence of the normalized text that contains a pay word. In 33.3 percent of the pairs in which the next advertisement mentions cash pay after a mention, at least one pay sentence of the first advertisement appears word for word in the next. The last two rows count a repeated mention only when no pay sentence is repeated. Masking the numbers in the sentences changes these shares little.

Table C2 repeats the consecutive-advertisement comparison for 2017–2019 and 2020–January 2022. Pairs are formed in the full posted-once sample as in Table 9, then retained when both dates fall in the relevant period. The transition rates are almost identical across the two periods.

For the prediction exercise in Table C4, the training sample consists of posted-once advertisements published before January 2021. We estimate disclosure rates separately by sector, position code, and employer, then use each set to predict the same advertisements published in 2021 and January 2022. To limit noise in small groups, a group with s disclosing advertisements out of n receives a predicted rate of $(s + 20\bar{p})/(n + 20)$, where $\bar{p}$ is the overall training-sample rate. Groups absent from the training sample receive $\bar{p}$.

Prediction quality is measured by the mean squared difference between the predicted probability and the outcome of zero or one, which is also called the Brier score. For cash pay, the rates of positions within sectors give a larger reduction in this error than the employer rates. For pay numbers and the minimum wage, and for non-wage benefits, the employer rates give the larger reduction. The rate of a position within a sector is shrunk towards the rate of the position, and the rate of the position towards the overall rate. The combined predictor adds to the rate of the position within the sector the employer's average deviation from this rate in the training sample, shrunk towards zero with the same weight of 20 advertisements. The combined predictor gives the largest reduction. The test includes 42,197 advertisements from employers absent from the training sample. For these advertisements, the employer

prediction defaults to the overall rate, and the combined predictor to the rate of the position within the sector.

Table C5 relates disclosure in an employer's first observed year to disclosure two or more calendar years later. Employers must have at least two posted-once advertisements in each period. Column (2) also includes the disclosure rate of other employers in the markets of the first-year advertisements, where a market is a position–province–year cell. The employer's own early rate is a stronger predictor of its later disclosure than the surrounding market rate. These are associations across employers, not estimates of learning from an initial market.

Disclosure and recruitment conditions

This part gives the details behind Section 8.3 and Table 11.

Inflation. Annual inflation moved between about 9 and 49 percent, while the monthly share of advertisements mentioning cash pay stayed between 1.7 and 3.2 percent (Figure C1). In a regression with employer effects, calendar-month-of-year effects and a linear trend, a 10-point rise in inflation is associated with a 0.08-point rise in cash-pay mentions (standard error 0.03). Without the trend, the estimate is 0.06 points (0.03). Numerical pay mentions rise slightly in these comparisons, and they do not fall. With the trend, a 10-point rise in inflation goes with a 0.054-point rise (0.019) in the share of advertisements that state a number or the minimum wage, against a mean of 0.53 percent. Non-wage benefit mentions increase by 0.19 points per 10 points of inflation (0.05).

Rare disclosure continued through a large rise in inflation, and it did not fall further. We cannot give this pattern a causal reading. Inflation affects all employers at once and coincides with other changes over time. Inflation could also discourage wage numbers that soon go out of date, and it could make current pay information more valuable. The estimates cannot isolate either force.

Market concentration. Employers with few rivals for recruits may have less reason to state pay. We measure concentration with a Herfindahl index of employer shares of advertisements in occupation–province–year cells. The index captures concentration among employers that post advertisements, which differs from concentration of employment. With employer, occupation and year effects, there is no clear relationship with disclosure. The unconditional disclosure rates are also close across concentration quartiles, between 2.5 and 2.7 percent.

Other employers' queues. Employers may disclose less when they expect many applications. In the raw data, pay mentions are more common where other employers' queues are shorter, at 3.3 percent in the lowest queue quartile and 2.1 percent in the highest. The measure uses other employers' advertisements in the same occupation, province and year. With

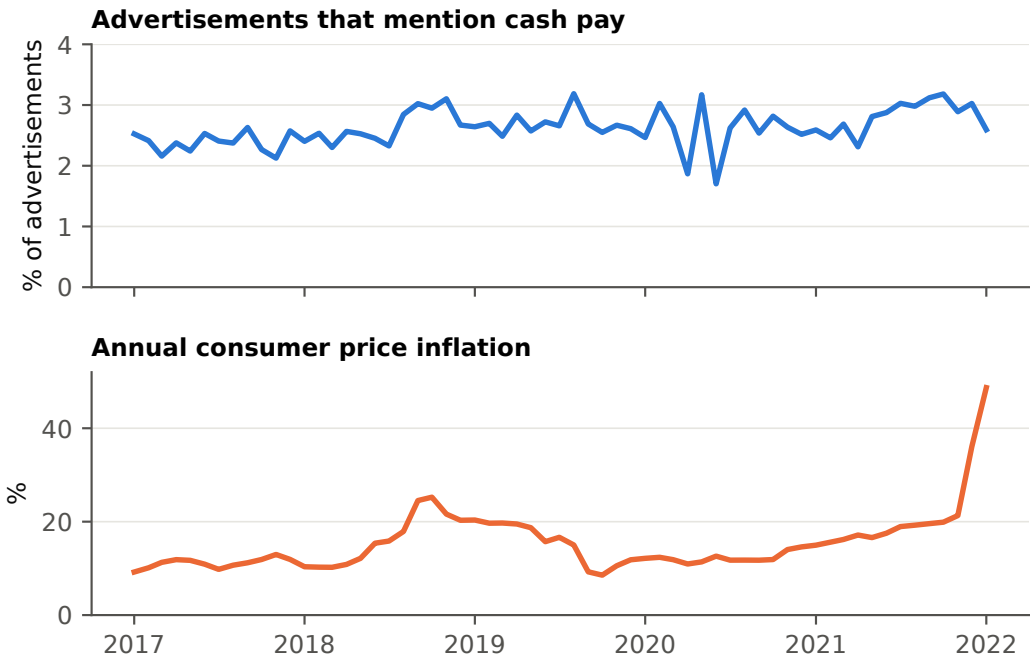


Figure C1. Pay disclosure and inflation, January 2017 to January 2022

Notes. The top panel shows the share of all advertisements first published in each month that mention cash pay. The bottom panel reports annual consumer price inflation from the Turkish Statistical Institute. The panels use different scales.

employer-by-year and occupation effects, doubling the queue measure goes with a change of −0.02 percentage points in disclosure (standard error 0.05). Since the conditional estimate is so small, much of the raw relationship probably reflects which employers and jobs are found in those markets.

Competing advertisements. We also count posted-once advertisements from other employers for the same position code, province and month. Disclosure is 2.6 percent in each of the two quartiles with fewer competing advertisements, and between 3.0 and 3.1 percent in the two higher quartiles. Doubling the measure goes with 0.03 more percentage points of disclosure when employer and month effects are included (0.02), and with 0.04 points when position effects are added (0.03). Neither estimate is statistically significant at the 5 percent level. The size is modest. Moving from the lowest quartile's median of one competing advertisement to the highest quartile's median of 38 corresponds to about 0.1–0.2 points, against a mean disclosure rate of 2.8 percent.

The employer's own experience. Among the 6,248 employers that ever report sufficient applications, disclosure after the first such report differs from disclosure before it by −0.08 points (0.12). Among newly observed employers, the within-employer association between the number of earlier advertisements and disclosure is also small and imprecise. These

comparisons cannot identify learning. Employers may have recruited elsewhere, or received enough applications without recording that reason. We simply find no clear sign that disclosure changes after the experiences we can observe.

Size of the associations. Table C6 gives each association in Panel A of Table 11 as the change over the interquartile range of the condition, in percentage points and as a percentage of the mean disclosure rate, with 95 percent confidence intervals. The table also reports the same changes for pay numbers and the minimum wage. Over the interquartile range of inflation, cash-pay mentions rise by 2.3 percent of the mean, with an interval from 0.6 to 3.9 percent. For market concentration and other employers' queues, the intervals for cash pay rule out changes larger than about 3 to 5 percent of the mean. For pay numbers and the minimum wage, the intervals are wider, because these mentions are rare.

Table C1. Pay disclosure by year

Year	Ads	Cash pay (%)	Number (%)	Number or minimum wage (%)	Benefit (%)
2017	180,679	2.38	0.32	0.39	3.22
2018	182,870	2.61	0.44	0.53	3.61
2019	165,132	2.69	0.49	0.60	3.80
2020	144,835	2.59	0.45	0.56	3.70
2021	274,559	2.86	0.44	0.56	4.31
January 2022	24,291	2.59	0.39	0.53	4.64

Notes. All advertisements, by date of first publication. Annual CPI inflation averaged 11 percent in 2017, peaked at 25 percent in October 2018, and reached 36 percent in December 2021 and 48.7 percent in January 2022.

Table C2. Persistence within each period

Period	Pairs	Previous ad did not (%)	Previous ad did (%)	Difference (SE)
2017–2019	300,699	1.37	44.61	43.25 (1.48)
2020–January 2022	229,895	1.36	44.74	43.37 (1.28)

Notes. Pairs are formed in the full posted-once sample as in Table 9 and kept when both dates fall in the period. Standard errors are clustered by employer.

Table C3. Persistence by type of pay information

First and second type	Pairs after a mention	Next ad has the second type (%): After a mention	After none	Difference (SE)
Same employer, different dates				
Any cash pay	13,086	44.2	1.39	42.8 (1.1)
Pay number or minimum wage	2,440	39.3	0.29	39.0 (1.8)
Pay number only	2,026	41.9	0.23	41.7 (2.1)
Other cash-pay language	10,646	39.3	1.20	38.1 (1.0)
Other language → number or minimum wage	10,646	2.9	0.41	2.5 (0.5)
Number or minimum wage → other language	2,440	13.3	1.88	11.4 (1.6)
Same recurring recruitment, different dates				
Any cash pay	6,819	79.1	0.68	78.4 (1.2)
Pay number or minimum wage	1,341	68.0	0.19	67.8 (2.7)
Pay number only	1,184	68.5	0.17	68.3 (3.0)
Other cash-pay language	5,478	76.0	0.60	75.4 (1.2)
Other language → number or minimum wage	5,478	2.8	0.50	2.3 (0.6)
Number or minimum wage → other language	1,341	12.1	2.16	9.9 (2.4)
Same employer: next ad mentions cash pay without repeating a pay sentence				
No pay sentence repeated word for word	13,086	29.5	1.39	28.1 (1.1)
Same, with numbers masked	13,086	29.0	1.39	27.6 (1.1)

Notes. Consecutive pairs of posted-once advertisements published on different dates, as in Table 9. Pairs after a mention count pairs whose first advertisement has the first type. Other cash-pay language is a cash-pay mention without a number or a reference to the minimum wage. The rows with an arrow give the probability that the next advertisement has the second type after the first type, and after an advertisement without the first type. A pay sentence is a sentence of the normalized text that contains a pay word. The last two rows count a pair as a repeat when the next advertisement mentions cash pay and none of its pay sentences appears word for word in the first advertisement. In 33.3 percent of the pairs in which the next advertisement mentions cash pay after a mention, at least one pay sentence is repeated. Standard errors clustered by employer are in parentheses.

Table C4. Predicting disclosure in 2021 and January 2022 from earlier advertisements

Outcome	Predictor	Mean squared error	Reduction (%)	Unseen test ads
Cash pay	Training mean	0.02870	0.00	0
Cash pay	Sector	0.02788	2.86	0
Cash pay	Position code	0.02607	9.16	1,735
Cash pay	Employer	0.02616	8.84	42,197
Cash pay	Position within sector	0.02539	11.52	
Cash pay	Position within sector, plus employer	0.02371	17.37	42,197
Non-wage benefit	Training mean	0.04223	0.00	0
Non-wage benefit	Sector	0.04184	0.93	0
Non-wage benefit	Position code	0.04076	3.47	1,735
Non-wage benefit	Employer	0.03730	11.66	42,197
Non-wage benefit	Position within sector	0.03998	5.32	
Non-wage benefit	Position within sector, plus employer	0.03602	14.71	42,197
Number or minimum wage	Position within sector	0.00586	2.68	
Number or minimum wage	Employer	0.00575	4.51	
Number or minimum wage	Position within sector, plus employer	0.00566	6.01	

Notes. Means are estimated on 583,365 posted-once advertisements before January 2021 and tested on the same 245,034 later advertisements for every predictor. Group means are shrunk towards the training mean with a weight of 20 advertisements, and unseen groups get the training mean. Position within sector shrinks the mean of each position-sector cell towards the position mean. Plus employer adds to that prediction the employer's average deviation from it in the training data, shrunk with the same weight. The reduction is relative to the training mean. The mean squared error is also called the Brier score. Lower is better. For cash pay, the employer rates give a mean squared error of 0.026162 and the rates of positions within sectors 0.025392.

Table C5. First-year disclosure and disclosure two or more years later

Sample	(1) First-year rate	(1) Employers	(2) First-year rate	(2) Market norm	(2) Employers
All employers	0.525	13,617	0.509	0.203	13,367
	(0.025)		(0.025)	(0.040)	
New employers (first seen in 2018 or later)	0.563	3,227	0.553	0.124	3,147
	(0.045)		(0.046)	(0.072)	

Notes. One observation per employer. The outcome is the employer's cash-pay rate two or more calendar years after its first observed year. Both periods need at least two posted-once advertisements. The market norm is the cash-pay rate of other employers in the same position, province and year, averaged over the employer's first-year advertisements. Column (2) drops employers that were the only advertiser in all of their first-year markets. Robust standard errors in parentheses. For all employers, the first observed year can be later than the true first year on the platform, because the data start in 2017.

Table C6. Disclosure over an interquartile change in each market condition

Condition	Interquartile range	Cash pay: Points	Cash pay: % of mean	Number or minimum wage: Points	Number or minimum wage: % of mean
Annual inflation (percent)	11.75 to 19.25	0.060	2.3	0.041	7.7
Effects: employer, month of year, trend		(0.017, 0.103)	(0.6, 3.9)	(0.013, 0.068)	(2.4, 13.0)
Employer concentration (HHI)	0.003 to 0.027	0.015	0.6	−0.027	−5.1
Effects: employer, occupation, year		(−0.039, 0.068)	(−1.5, 2.6)	(−0.059, 0.005)	(−11.1, 0.9)
Other employers' queues (applications)	122 to 311	−0.027	−1.0	−0.045	−7.9
Effects: employer × year, occupation		(−0.147, 0.093)	(−5.4, 3.4)	(−0.106, 0.016)	(−18.6, 2.8)
Competing advertisements	2 to 16	0.077	2.7	0.045	7.4
Effects: employer, month		(−0.028, 0.182)	(−1.0, 6.5)	(−0.002, 0.091)	(−0.3, 15.1)
Competing advertisements	2 to 16	0.119	4.2	0.040	6.6
Effects: employer, position, month		(−0.037, 0.276)	(−1.3, 9.8)	(−0.061, 0.141)	(−10.2, 23.4)

Notes. Each row is one regression of the outcome (100 if the advertisement has the type of pay information, 0 otherwise) on the condition, with the fixed effects named below it, as in Panel A of Table 11. The table reports the coefficient times the interquartile range of the condition across the advertisements of the regression, in percentage points and as a percentage of the mean outcome of the regression (cash pay 2.65 to 2.82 percent, number or minimum wage 0.53 to 0.60 percent). Inflation is in percent per year. Queues are the geometric mean of one plus the applications to other employers' advertisements in the occupation, province and year. Competing advertisements are those of other employers for the same position in the same province and month. The 95 percent confidence intervals in parentheses use standard errors clustered as in Table 11, by calendar month for inflation and by employer otherwise.

D Closure Reasons and Posting Windows

Table D1 groups the recorded closure reasons. Candidate-found reports outside Kariyer.net include referrals, existing staff, the employer's website, other job sites, consultancies, newspaper advertisements, and other sources. We combine these into one category, combine frozen and canceled positions, and pool spelling variants of each reason. No recorded reason is not evidence that recruitment failed.

Table D2 reports the span between first publication and the recorded end date. A 59-day difference covers 60 calendar dates when both endpoints are counted. We therefore group spans of 59 and 60 days as 60-day packages and spans of 29 and 30 days as 30-day packages. These spans largely describe the package, not vacancy duration. Table D3 reports the final queues by package. Advertisements with a 60-day package receive more applications, with a median of 285 against 143 for 30-day packages. Even the 30-day packages have a mean queue of 234 applications.

Table D4 reports both the probability of recording any reason and the mix of reasons conditional on reporting. Recording is most common in the shortest queues, where frozen or canceled positions account for most recorded reasons. Neither the unconditional recording rate nor the conditional reason shares identify hiring probabilities.

Pay information and closure reports

In the main specification, ads that state pay get more applications, but the size of this link is not stable (Section 6.2). More applications also do not necessarily translate into better recruitment outcomes. Here we examine whether pay information also accompanies the reasons that employers record when closing an ad (Section 3.2.1).

We regress four closure outcomes on the indicators from equation (1). Three outcomes are equal to one if the employer records a specific reason. These reasons are that the employer (a) found a candidate via the platform, (b) found a candidate via any source, or (c) received sufficient applications. An ad without any reason recorded has a value of zero. Therefore, each of these outcomes captures whether the employer chose to select the reason, not whether that reason was the true underlying cause of the closure. The fourth outcome equals one if the employer records any reason whatsoever, indicating whether pay information is associated with a general tendency to report reasons. The key variables of interest are cash pay stated with and without a figure, each relative to a baseline where cash pay is not mentioned. Similar to Table 7, these regressions incorporate recurring-recruitment and market-month fixed effects along with identical control variables. Estimates are provided in Table D5.

Findings on finding a candidate through the platform are positive but estimated imprecisely. Stating cash pay without specifying a number is correlated with a 0.68 percentage point increase in the likelihood that the employer reports finding a candidate (standard error 0.83), compared to a baseline average of 9.34 percent. For ads stating a specific pay figure, the coefficient is 1.32 points (SE = 1.53). With cash pay mentioned but no number given, the minimum detectable effect at 80 percent power and a 5 percent significance level is 2.35 points, roughly one-quarter of the mean. Consequently, we cannot exclude the possibility that pay disclosure is associated with identifying candidates through the platform. Furthermore, mentioning cash pay without a figure is correlated with a 1.48 point rise in the likelihood that the employer documents any reason at all (standard error 1.17), which is not statistically significant either. Employers who record a reason name a candidate found through the platform in 38 percent of cases (9.34 of 24.36 percent). If the additional reports followed this pattern, they alone would raise the probability of reporting a candidate found through the platform by about 0.57 points (1.48 times 0.383). This number is close to the estimate of 0.68 points. So the higher rate of such reports may only reflect that these employers fill in the closure form more often, and not that pay disclosure helps them hire. Because we lack data on whether the employer genuinely found a candidate via the platform, distinguishing between these explanations is not possible.

Where do all applications sit?

The sufficiency distribution in Section 7.1 gives two benchmarks that we can apply to all posted-once advertisements, its lower quartile of 108 applications and its median of 250. Table D6 reports two shares for each benchmark c. The first counts all applications in advertisements whose final queue exceeds c. The second counts only the applications above c in each such advertisement. With final queue A_i, the two shares are

$$S(c) = \frac{\sum_i A_i \, 1\{A_i > c\}}{\sum_i A_i} \qquad \text{and} \qquad E(c) = \frac{\sum_i \max(A_i - c, 0)}{\sum_i A_i}. \tag{D1}$$

Of all posted-once advertisements, 71.7 percent have queues above 108. These advertisements hold 95.9 percent of all applications, and the applications above 108 in each of them add up to 74.6 percent. At 250, the corresponding shares are 44.1, 82.8 and 52.5 percent. Most applications sit in long queues, and about half lie above the median final queue of advertisements with a sufficiency report.

Neither share measures wasted effort, for three reasons. a) The benchmarks come from a selected group of employer reports. b) Openings and hiring needs differ across advertisements. c) We do not observe when each application arrived. The shares are simple arithmetic on

final counts and do not estimate how many applications came in after an employer stopped needing them. Appendix F shows how the two shares are related.

Table D1. Reasons employers record when closing an advertisement

	All advertisements		Posted once	
Recorded reason	N	%	N	%
Candidate through platform	86,053	8.85	68,030	8.21
Candidate through other source	33,228	3.42	26,207	3.16
Sufficient applications	38,691	3.98	29,064	3.51
Frozen or canceled	68,364	7.03	52,074	6.29
No candidate found	19	0.00	15	0.00
No recorded reason	746,011	76.72	653,009	78.83

Notes. Spelling variants of the same reason are combined. Recording a reason is voluntary. No recorded reason does not mean that recruitment failed.

Table D2. Recorded spans from first publication to the end date

Recorded span	Advertisements	%
0 days	139	0.02
29 days	228,158	27.54
30 days	7,289	0.88
59 days	397,743	48.01
60 days	69,717	8.42
Any other span	125,353	15.13

Notes. Posted-once advertisements, N = 828,399. A span of 59 days covers 60 calendar dates if both ends are counted. The spans are recorded windows. They do not measure how long an advertisement was active or how long it took to fill the job.

Table D3. Applications by posting package

			Applications						
Recorded span	Ads	%	Mean	P25	Median	P75	P90	% with none	Median views
30-day package (29 or 30 days)	235,447	28.4	234	68	143	282	505	1.9	1,610
60-day package (59 or 60 days)	467,460	56.4	460	140	285	542	945	1.5	2,401
Other recorded spans	125,492	15.1	253	51	139	306	571	2.5	1,519
All posted-once	828,399	100.0	365	95	214	431	780	1.8	2,007

Notes. Posted-once advertisements. The recorded span runs from first publication to the recorded end date (Appendix Table D2). The span shows the package that the employer bought, not how long the advertisement was actually visible.

Table D4. Recorded reasons across final queues

Final queue	Ads	Any reason recorded (%)	Among advertisements with a reason (%)			
			Platform	Other source	Sufficient	Frozen
0–24	56,220	29.2	9.4	12.1	12.0	66.5
25–74	106,001	17.9	27.5	17.0	16.3	39.2
75–149	150,301	18.8	36.1	16.8	16.3	30.8
150–249	149,047	20.1	41.9	15.7	16.2	26.1
250–499	196,554	21.5	46.1	15.0	16.4	22.5
500–999	118,625	22.9	49.2	13.7	17.2	19.8
1,000+	51,651	24.0	46.0	11.9	23.5	18.6

Notes. Posted-once advertisements. The last four columns are compositions of the reasons that employers chose to record. They are not hiring probabilities.

Table D5. Pay information and what employers record at closure

	Candidate found on the platform	Candidate found (any source)	Sufficient applications	Any reason recorded
Cash pay without a number	0.68	0.32	−0.91	1.48
	(0.83)	(0.93)	(0.81)	(1.17)
Cash pay with a number	1.32	0.77	−0.86	1.49
	(1.53)	(1.71)	(1.20)	(2.48)
Non-wage benefit	−0.45	0.05	−0.43	−1.47
	(0.66)	(0.79)	(0.51)	(1.01)
Outcome mean (%)	9.34	12.29	4.37	24.36
MDE, cash pay without a number	2.35	2.65	2.30	3.32
Advertisements		225,887		

Notes. Outcomes and coefficients are in percentage points. Each column is one regression on the common sample, with the effects and controls of Table 6. An advertisement without a recorded reason counts as zero, so each outcome is the probability that the employer records that reason. None is a verified hire. MDE is the minimum detectable effect with 80 percent power at the 5 percent level, $(t_{0.975,60} + z_{0.80})$ times the standard error.

Table D6. Applications above two sufficiency benchmarks

Benchmark	Advertisements with $A_i > c$	Applications in these advertisements, $S(c)$	Applications above c, $E(c)$
Lower quartile, $c = 108$	71.7	95.9	74.6
Median, $c = 250$	44.1	82.8	52.5

Notes. All posted-once advertisements. Entries are percentages. The benchmarks are the lower quartile and the median of final queues among the 29,064 posted-once advertisements with a sufficiency report. A_i is the final queue of advertisement i. $S(c)$ is the share of all applications that belong to advertisements with $A_i > c$. $E(c)$ is the share of all applications that lie above c within those advertisements. Neither share measures wasted effort.

E Disclosure by Sector and Position Level

Table E1 ranks the 78 sectors with at least 2,000 advertisements by the share mentioning cash pay. Call centers, real estate, beauty and wellness, advertising, law firms, e-commerce, and security are among the highest. Security also has the highest numerical-pay share, at 4.4 percent. Holding companies, audit services, cement, hospitals, kitchenware, and hosiery are among the lowest.

Table E2 uses the position level selected from the platform menu. Freelance advertisements have the highest cash-pay rate, 16.1 percent, but represent a small group. Among the larger categories, staff and new-starter jobs disclose more often than specialist and manager jobs. The final row identifies lower-level jobs that mention high school but not a university degree, used as a proxy for jobs more likely to pay near the statutory minimum. It does not classify observed wages. In this group, 0.43 percent state a number and 0.21 percent name the minimum wage; allowing for overlap, 0.61 percent provide either form of information.

Table E1. Pay disclosure by employer sector

Sector	Ads	Employers	Cash pay (%)	Number (%)
Highest disclosure				
Call centers	5,390	317	19.96	4.36
Real estate	10,296	1,961	19.19	1.29
Beauty and wellness centers	2,813	1,005	13.12	1.92
Advertising and promotion	4,989	1,060	10.86	2.55
Law firms	3,541	843	9.97	0.59
E-commerce	2,557	225	8.68	1.96
Other	17,223	7,650	8.36	1.81
Security	5,530	422	7.59	4.41
Insurance	8,696	1,037	6.61	0.31
Services	20,449	2,764	6.31	1.63
Lowest disclosure				
Holding companies	32,960	186	0.45	0.06
Audit services	2,622	149	0.38	0.04
Cement	2,435	51	0.37	0.08
Hospitals	2,542	108	0.35	0.04
Kitchenware	2,507	64	0.20	0.00
Hosiery	4,654	10	0.02	0.00

Notes. All advertisements. Only the 78 sectors with at least 2,000 advertisements are ranked. Sector names are translated from the platform's Turkish labels.

Table E2. Pay disclosure by position level

Position level	Ads	Cash pay (%)	Number (%)	Minimum wage (%)
Specialist	326,857	1.96	0.22	0.03
Not recorded	259,395	3.45	0.65	0.15
Staff	117,753	4.07	0.81	0.29
Middle manager	87,025	1.13	0.10	0.01
Blue collar	36,877	1.94	0.30	0.22
Assistant specialist	36,453	1.56	0.16	0.02
New starter	34,373	3.69	0.47	0.20
Service staff	26,438	2.92	0.91	0.26
Management trainee	23,950	2.09	0.27	0.02
Senior manager	13,581	2.17	0.18	0.01
Intern	7,373	1.71	0.11	0.20
Freelance	2,291	16.11	1.53	0.26
Jobs likely to pay the minimum wage	53,482	2.64	0.43	0.21

Notes. All advertisements. Position levels are the platform's structured field, translated. Jobs likely to pay the minimum wage are new-starter, staff, intern, blue-collar or service-staff jobs that mention high school and do not mention a university degree. In these jobs, 0.61 percent of advertisements state a number or name the minimum wage, compared with 0.53 percent of all advertisements.

F Two Identities

Persistence when employers differ

Consider consecutive advertisements from employer f, which mentions pay with fixed probability p_f, independently across advertisements. Let w_f be its share of all pairs, $\bar{p} = \sum_f w_f p_f$ the pair-weighted mean, and $V = \sum_f w_f (p_f - \bar{p})^2$ the corresponding variance. Then

$$\Pr(D_t = 1 \mid D_{t-1} = 1) = \frac{\sum_f w_f p_f^2}{\bar{p}} = \frac{\bar{p}^2 + V}{\bar{p}} = \bar{p} + \frac{V}{\bar{p}}, \tag{F1}$$

$$\Pr(D_t = 1 \mid D_{t-1} = 0) = \frac{\sum_f w_f p_f (1 - p_f)}{1 - \bar{p}} = \frac{\bar{p}(1 - \bar{p}) - V}{1 - \bar{p}} = \bar{p} - \frac{V}{1 - \bar{p}}. \tag{F2}$$

Subtracting gives

$$\frac{V}{\bar{p}} + \frac{V}{1 - \bar{p}} = \frac{V}{\bar{p}(1 - \bar{p})}, \tag{F3}$$

which is equation (2). In the 554,515 employer-level pairs, the previous advertisement mentions cash pay in 2.36 percent of cases. Taking $\bar{p} = 0.0236$, a transition gap of 0.428 implies $V = 0.0099$, or a standard deviation of 0.099. Under this illustrative model, a standard deviation of about 10 percentage points across employer disclosure probabilities

could generate the observed gap without one disclosure affecting the next. This calculation does not estimate the actual distribution of p_f.

Two shares of applications

For a benchmark queue c, $S(c)$ counts all applications in advertisements with $A_i > c$, while $E(c)$ counts $A_i - c$ in each such advertisement. The difference is therefore

$$S(c) - E(c) = \frac{\sum_i c\, 1\{A_i > c\}}{\sum_i A_i} = \frac{c\ \Pr(A > c)}{\mathrm{E}[A]}, \tag{F4}$$

where the probability and mean are taken over advertisements. At $c = 250$, 44.1 percent of posted-once advertisements exceed the benchmark and the mean queue is 364.5. Thus $250 \times 0.441/364.5 = 0.30$ after rounding, consistent with the difference between 82.8 and 52.5 percent in Table D6. The verification code checks this identity for each benchmark. It is an identity about final counts, not the timing or value of the applications.